# Multi-Impairment Stability Bounds for Smart Grid Communication: Learnable Delay, Packet Loss, and Jitter Coupling

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***Abstract*—Existing stability bounds for smart grid communication account only for delay, treating the channel as lossless and jitter-free. This paper shows that delay-only models yield a margin gap of 11–25% relative to a full multi-impairment assessment, widening to 24–25% for grids exceeding 1,000 buses. We derive a stability bound decomposing margin erosion into three learnable terms for delay, packet loss, and jitter, with per-generator coupling constants trained end-to-end. The proposed ResilientJointOptimizer combines dual-domain graph neural network encoders, physics-constrained attention, and a three-state Markov channel predictor with budget-aware initialization informed by the empirical dominance of packet loss over jitter. Evaluation across eight power systems (39 to 2,869 buses), seven baselines, and 20 seeds confirms statistical significance at all conventional levels. Under severe degradation, delay-only models declare full stability while the proposed method identifies instability, yielding a false-safe rate up to 100% for the largest grids. Per-generator erosion follows a power-law scaling, and parameter overhead remains below 4%, supporting practical deployment.**

***Index Terms*—Smart grid communications, stability margin, packet loss, jitter, graph neural networks, multi-impairment bound.**

## I. Introduction

THE reliable operation of bulk power systems is increasingly contingent upon the integrity of the communication channels that carry measurement telemetry and control setpoints between substations, control centers, and distributed energy resources [1]–[3]. Field experience has repeatedly confirmed that communication degradation, rather than generation shortfall, is the proximate trigger of wide-area disturbances. During the February 2021 Texas grid crisis, the loss of SCADA visibility at multiple gas-fired plants, caused by frozen communication equipment and congested IP backhauls, prevented operators from issuing timely dispatch corrections, contributing to 20 GW of involuntary load shedding across ERCOT [4]. At smaller scale, a 2019 cyberattack on a Western US utility disrupted firewalls between the SCADA network and the corporate WAN for over 12 hours, intermittently dropping PMU packets and injecting anomalous jitter into automatic generation control (AGC) signals [5]. In both incidents, the salient impairment was not merely latency but the compound effect of packet loss, delay fluctuation, and sustained throughput degradation acting simultaneously on the control loop. Three categories of channel impairment co-exist in every realistic wide-area network: *delay* $\tau$, the end-to-end latency of control packets; *packet loss rate* $p$, the fraction of measurement or setpoint packets that fail to arrive; and *jitter* $\sigma_j$, the standard deviation of inter-packet arrival time variation that causes irregular sampling at the controller. Despite this empirical reality, every formal stability bound published to date for communication-dependent power systems

TABLE I
Summary of Key Notation

| Symbol | Description |
|---|---|
| *Power system dynamics* | |
| $\mathcal{G}_E, \mathcal{G}_I$ | Power grid and communication graphs |
| $N, n_g$ | Number of buses and generators |
| $\boldsymbol{\theta}(t)$ | Rotor angle vector |
| $\boldsymbol{\Gamma}, \boldsymbol{\Phi}$ | Generator inertia and damping matrices |
| $\mathbf{S}$ | Synchronizing torque matrix |
| $\mathbf{J}(0)$ | System matrix at zero impairment |
| $\mathbf{z}(t)$ | Deviation state vector $[\Delta\boldsymbol{\theta}^\top, \Delta\dot{\boldsymbol{\theta}}^\top]^\top$ |
| $\mathbf{B}$ | Control input matrix |
| $\mathbf{u}$ | Generator dispatch setpoints |
| $\lambda_{\min}(0)$ | Least stable eigenvalue at zero impairment |
| $\mathbf{V}, \mathrm{cond}(\mathbf{V})$ | Eigenvector matrix and its condition number |
| *Communication impairments* | |
| $\tau_i, p_i, \sigma_{j,i}$ | Per-generator delay, packet loss rate, jitter |
| $\mathbf{x}_I \in \mathbb{R}^{N\times 6}$ | Comm. input: $[\tau, p, \sigma_j, R_{\mathrm{retx}}, B, \mathrm{SNR}]$ |
| $\boldsymbol{\Delta}_{\mathrm{loss}}, \boldsymbol{\Delta}_{\mathrm{jitter}}$ | Packet-loss and jitter perturbation terms |
| *Stability bound (Theorem 2)* | |
| $K_i = \exp(\kappa_i)$ | Delay coupling constant for generator $i$ |
| $R_i = \exp(\varrho_i)$ | Packet-loss coupling constant |
| $J_i = \exp(\zeta_i)$ | Jitter coupling constant |
| $\Delta_K, \Delta_R, \Delta_J$ | Delay, packet-loss, and jitter erosion terms |
| $\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j)$ | Multi-impairment stability margin |
| $\rho_{\min}$ | Minimum stability threshold |
| $s$ | Safety factor for budget-aware initialization |
| *Architecture and training* | |
| $\mathbf{x}_E \in \mathbb{R}^{N\times 5}$ | Energy input: $[P, Q, V, \theta, \omega]$ |
| $\mathbf{h}_E, \mathbf{h}_I$ | Domain-specific GNN embeddings |
| $\mathbf{h}_{\mathrm{fused}}$ | Fused representation after attention |
| $d, H, L$ | Embedding dim, heads, GNN layers |
| $\mathbf{M}_{\mathrm{phys}}, \mathbf{M}_{\mathrm{causal}}$ | Physics and causal attention masks |
| $\mathcal{L}_{\mathrm{OPF}}, \mathcal{L}_{\mathrm{QoS}}$ | Energy and communication losses |
| $\mathcal{L}_{\mathrm{stab}}, \mathcal{L}_{\mathrm{couple}}$ | Stability hinge, coupling alignment losses |
| $\mathcal{L}_{\mathrm{channel}}$ | Channel state prediction loss |
| $\alpha, \beta$ | Coupling and channel loss weights |

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accounts only for delay, implicitly assuming lossless, jitter-free channels [6], [7]. This paper shows that such delay-only bounds yield a margin gap of 11—25% relative to a full multi-impairment assessment, a difference that grows with system size and may lead operators to underappreciate the cumulative effect of packet loss and jitter on the true stability boundary.

Four strands of prior research bear on this problem, and each leaves a critical gap that the present work addresses. *(i) Delay-dependent stability theory.* Lyapunov-Krasovskii functionals and linear matrix inequality (LMI) techniques provide the most rigorous certificates for time-delay systems [6]–[8]. Their coupling constants are derived analytically from system models, yielding verifiable delay margins. Yet every such formulation sets packet loss rate and jitter to zero by construction; the resulting margin does not account for loss- or jitter-induced erosion, leaving a gap that widens as channel conditions degrade. *(ii) Communication-aware power system optimization.* Robust OPF formulations based on minimax uncertainty sets [9] and stochastic OPF via scenario sampling [10] model generation and load uncertainty but subsume communication impairments into generic scenario distributions rather than treating delay, loss, and jitter as distinct physical erosion mechanisms with independent coupling constants. At the network edge, IoT-enabled architectures jointly optimize communication and computation resources for energy systems [11], graph reinforcement learning methods allocate IoT resources under time-varying topologies [12], high-reliability low-latency routing protocols address communication quality in cooperative systems [13], and deterministic scheduling methods target ultralow-latency communication in edge-enabled IIoT [14], yet these formulations focus on throughput, scheduling, or routing rather than closed-loop stability certificates under impaired channels. *(iii) Data-driven dispatch and topology-aware learning.* Neural-network-based OPF surrogates [15], [16] and GNN-based power flow solvers [17], [18] achieve fast inference and topology generalization, while complementary deep learning methods address adjacent tasks including attention-based load forecasting [19], [20], temporal pattern modeling [21], and LLM-based spectral time series analysis [22]. Secure communication frameworks further address data integrity under heterogeneous conditions [23]. Yet none of these architectures incorporate a formal stability bound tied to communication channel quality; the network infrastructure is treated as transparent. *(iv) Our predecessor delay-only framework.* The JointOptimizer of [24] was the first to learn per-generator delay-stability coupling constants $K_i$ end-to-end and was validated on five standard test cases (14, 30, 39, 57, and 118 buses). That work acknowledged, however, that the single-term bound becomes tight at larger scales: on the 118-bus case the delay-only erosion $\sum_i K_i\tau_i/\tau_{\max,i}$ consumes most of the available eigenvalue budget, leaving limited headroom for N-1 contingencies or non-lognormal delay tails [24]. The present paper addresses this limitation by extending the bound with two additional erosion terms $\Delta_R$ and $\Delta_J$ and distributing the eigenvalue budget across three impairment families via a budget-aware initialization, enabling the ResilientJointOptimizer to maintain positive margins from Case-39 through PEGASE-2869 (510 generators). To the best of our knowledge, no existing work jointly bounds delay, packet loss, and jitter stability margins with learnable per-generator coupling constants, nor evaluates such bounds beyond 118-bus systems.

To bridge this gap, this paper introduces the *ResilientJointOptimizer*, extending the delay-only JointOptimizer of [24] with three coordinated contributions. The theoretical foundation is Theorem 2, showing that $\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j) \geq |\lambda_{\min}(0)| - \Delta_K - \Delta_R - \Delta_J$, where $\Delta_K$ is linear in normalized delay (inherited from [24]), $\Delta_R$ is proportional to $p_i/(1-p_i)$ from the geometric distribution of Bernoulli drop runs, and $\Delta_J$ is quadratic in jitter standard deviation from a second-order Taylor expansion. Three families of per-generator coupling constants $K_i$, $R_i$, $J_i$ are constrained positive via the exponential map and learned end-to-end, replacing fixed worst-case analytical constants with data-driven values. A budget-aware initialization allocates 50% of the eigenvalue budget to delay, 30% to packet loss, and 20% to jitter, reflecting the empirical $\sim 88/12$ dominance of packet loss over jitter. A three-state Markov channel predictor grounds the communication embeddings in physically meaningful dynamics and prevents the loss and jitter constants from collapsing to zero.

Building on this architecture, the contributions of this paper are fivefold:

**C**1. We prove Theorem 2, a multi-impairment stability bound $\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j) \geq |\lambda_{\min}(0)| - \Delta_K - \Delta_R - \Delta_J$, extending the delay-only Theorem 1 of [24] to jointly capture packet-loss erosion via $R_i \cdot p_i/(1-p_i)$ and jitter erosion via $J_i \cdot \sigma_{j,i}^2/\sigma_{\max,i}^2$. The proof derives the packet-loss term from the geometric distribution of Bernoulli drop runs and the jitter term from a second-order Taylor expansion, then assembles all three via the Bauer-Fike eigenvalue perturbation theorem and the triangle inequality.

**C**2. We introduce learnable per-generator coupling constants $K_i$, $R_i$, $J_i$ that replace worst-case fixed theoretical values with data-driven estimates trained end-to-end via gradient descent. Ablation analysis reveals that packet loss accounts for 51—58% of total erosion and jitter for 6—7.4%, with delay comprising the remainder (35—42%), providing the first quantitative decomposition of multi-impairment margin erosion by source.

**C**3. We quantify the margin gap between delay-only and multi-impairment bounds at 11—25%, with the difference ranging from 11.1—11.5% on small grids (Case-39 to Case-300) to 24.4—24.7% on bulk grids (PEGASE-1354 to PEGASE-2869). Under stressed impairments ($p = 0.30$, $\sigma_j = 80\,\mathrm{ms}$), the gap ratio grows to $3.6\times$.

**C**4. We discover a $N^{-0.55}$ erosion scaling law, showing that per-generator margin erosion decreases with grid size as individual generator impact is diluted, while aggregate risk grows with the number of generators. Comprehensive evaluation spans eight real power systems (39 to 2,869 buses, representing New England, US Midwest, Eastern US, European, and pan-European networks), seven comparison models, and 20 independent seeds with rigorous statistical validation (Wilcoxon signed-rank $p = 1.91 \times 10^{-6}$, Cohen's $d > 30$, Friedman $\chi^2 = 120$, $p = 1.63 \times 10^{-23}$).

**C5.** We demonstrate practical deployment viability: the proposed framework adds only 2.9—3.8% parameter overhead relative to the delay-only model, with inference latency below 60 ms for all eight cases and below 25 ms for cases up to 300 buses, establishing multi-impairment awareness as a practical deployment necessity.

The remainder of this paper is organized as follows. Section II presents the coupled power-communication system model and formulates the joint optimization problem with multi-impairment stability constraints. Section III develops the theoretical analysis, including the full statement and proof sketch of Theorem 2 and the erosion scaling law. Section IV describes the ResilientJointOptimizer architecture, the learnable coupling constant families, the Markov channel predictor, and the composite training objective. Section V details the experimental setup, covering the eight test systems, seven baseline methods, training protocol, and statistical methodology. Section VI presents the results and discussion. Section VII discusses limitations and practical deployment applications. Section VIII concludes. Fig. 2 illustrates the main performance comparison that motivates the need for multi-impairment awareness.

## II. System Model and Preliminaries

Building on the delay-only formulation of [24], which modeled the communication channel as a pure latency element, this section extends the coupled power-communication system model to incorporate three simultaneous impairment sources: delay, packet loss, and jitter. The swing equation is augmented with additive perturbation terms for dropped and irregularly timed control packets, producing a perturbed delay-differential equation whose multi-impairment stability properties are analyzed in Section III. The communication overlay is characterized by a six-dimensional feature vector per node—three more than the delay-rate-buffer triple of [24]—to capture the richer channel state required for joint packet-loss and jitter awareness. The joint optimization problem is then stated. Key notation appears in Table I.

### A. Power System Dynamics Under Communication Impairments

When dispatch commands and AGC corrections are delivered over realistic communication channels, three categories of degradation may simultaneously affect the control signal received by each generator: latency ($\tau_i$), packet loss ($p_i$), and arrival-time jitter ($\sigma_{j,i}$). The classical swing equation [25], [26], extended to account for these impairments, governs the aggregate rotor dynamics of $n_g$ synchronous generators as

$$\mathbf{\Gamma}\ddot{\boldsymbol{\theta}}(t) + \mathbf{\Phi}\dot{\boldsymbol{\theta}}(t) = \mathbf{P}_m(t-\boldsymbol{\tau}) - \mathbf{P}_e(\boldsymbol{\theta}(t)), \tag{1}$$

where $\boldsymbol{\theta}(t) \in \mathbb{R}^{n_g}$ is the vector of rotor angles; $\mathbf{\Gamma} = \mathrm{diag}(\Gamma_1, \ldots, \Gamma_{n_g})$ with $\Gamma_i = 2H_i/\omega_s$ collects the generator inertia constants; $\mathbf{\Phi} = \mathrm{diag}(\Phi_1, \ldots, \Phi_{n_g})$ gathers the mechanical damping coefficients; $\mathbf{P}_m(t-\boldsymbol{\tau})$ is the mechanical power command vector, each component arriving with channel-induced delay $\tau_i$ [27]; and $\mathbf{P}_e(\boldsymbol{\theta})$ are the electrical power injections governed by the network admittance matrix [28].

Small-signal linearization of (1) around a stable equilibrium $(\boldsymbol{\theta}_0, \dot{\boldsymbol{\theta}}_0 = \mathbf{0})$ with deviation state $\mathbf{z} = [\Delta\boldsymbol{\theta}^\top,\ \Delta\dot{\boldsymbol{\theta}}^\top]^\top \in \mathbb{R}^{2n_g}$ yields the multi-impairment state-space representation

$$\dot{\mathbf{z}}(t) = \mathbf{J}(0)\,\mathbf{z}(t) + \mathbf{B}\,\mathbf{u}(t-\boldsymbol{\tau}) + \boldsymbol{\Delta}_{\mathrm{loss}}(t) + \boldsymbol{\Delta}_{\mathrm{jitter}}(t), \tag{2}$$

where the nominal (zero-impairment) system matrix is

$$\mathbf{J}(0) = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{\Gamma}^{-1}\mathbf{S} & -\mathbf{\Gamma}^{-1}\mathbf{\Phi} \end{bmatrix} \in \mathbb{R}^{2n_g \times 2n_g},$$

with $\mathbf{S} = \partial\mathbf{P}_e/\partial\boldsymbol{\theta}|_{\boldsymbol{\theta}_0}$ being the synchronizing torque matrix derived from the network admittance and operating point, and $\mathbf{B} = [\mathbf{0};\ \mathbf{\Gamma}^{-1}]$ the control input matrix. Crucially, unlike the delay-only formulation in [24] where $\boldsymbol{\Delta}_{\mathrm{loss}} = \boldsymbol{\Delta}_{\mathrm{jitter}} = \mathbf{0}$, the present model retains two additive perturbation terms: $\boldsymbol{\Delta}_{\mathrm{loss}}(t)$ captures the sample-and-hold effect of dropped control packets, and $\boldsymbol{\Delta}_{\mathrm{jitter}}(t)$ captures the second-order variance from random arrival-time fluctuations. All eigenvalues of $\mathbf{J}(0)$ lie in the open left half-plane under nominal conditions [25]; Section III quantifies how the three impairment sources jointly shift these eigenvalues rightward and erode the stability margin.

The grid topology is represented as a graph $\mathcal{G}_E = (V, \mathcal{E}_E)$ with $N = |V|$ buses. Eight real benchmarks, spanning 39 to 2,869 buses (Table IX in Appendix B), cover regional (New England), continental (US Eastern Interconnection), and pan-European (PEGASE) transmission systems. The undelayed margin $|\lambda_{\min}(\mathbf{J}(0))|$ varies non-monotonically with system size, from 0.035 (Case-39) to 0.079 (PEGASE-2869), because generator inertia distributions and network connectivity differ across benchmarks. Larger grids enjoy a wider initial margin but accumulate proportionally greater aggregate erosion from their additional generators, producing the two-regime margin-gap structure analyzed in Section VI-C. Per-bus electrical features comprise active power $P$, reactive power $Q$, voltage magnitude $V$, voltage angle $\theta$, and frequency deviation $\omega$, forming $\mathbf{x}_E \in \mathbb{R}^{N\times 5}$.

### B. Communication Network Model with Impairments

A communication overlay graph $\mathcal{G}_I = (V, \mathcal{E}_I)$ is defined on the same bus set $V$ as the power grid; the edge set $\mathcal{E}_I$ may differ from $\mathcal{E}_E$ to model dedicated fiber or wireless links [29], [30]. All experiments adopt $\mathcal{E}_I = \mathcal{E}_E$ as a conservative lower bound on communication redundancy. A key departure from the predecessor framework [24], which described each node with only three communication features (delay $\tau$, data rate, buffer occupancy), is the adoption of a *six-dimensional* per-node feature vector $\mathbf{x}_I = [\tau,\ p,\ \sigma_j,\ R_{\mathrm{retx}},\ B,\ \mathrm{SNR}]^\top \in \mathbb{R}^6$. Here $\tau$ is the one-way end-to-end latency (ms), $p$ is the packet loss rate, $\sigma_j$ is the delay jitter standard deviation (ms), $R_{\mathrm{retx}}$ is the TCP-equivalent retransmission count, $B$ is the available buffer depth (packets), and SNR is the link signal-to-noise ratio (dB). The three additional features—$p$, $\sigma_j$, and SNR—are essential for the multi-impairment stability bound of Theorem 2 and for the channel-state predictor of Section IV-C. Providing all six features to every compared model ensures that performance differences reflect the stability-bound formulation, not unequal input information.

Three stochastic impairment sources are modeled as follows. Delays follow a truncated lognormal distribution calibrated to empirical SCADA and PMU latency surveys [29], given by

$$\tau_i \sim \mathrm{LogNormal}(\mu = 50\,\mathrm{ms},\ \sigma = 20\,\mathrm{ms}), \quad \tau_i \in [5,\ 500]\,\mathrm{ms}, \tag{3}$$

The 5 ms floor corresponds to fiber-optic propagation minima, while the 500 ms ceiling matches the maximum tolerable SCADA control-loop latency specified in IEEE C37.118.1 [31]. The delay distribution in (3) is consistent with empirical measurements reported in [29]. Packet loss rates are sampled from a Beta distribution with mean 5% over $[0, 30\%]$, representing independent Bernoulli packet drops in congested or interference-prone network segments. Upon a dropped packet at generator $i$, the controller holds the most recent successfully received setpoint; consecutive drops before a successful reception are geometrically distributed with mean $p_i/(1-p_i)$, yielding an effective hold-on delay of $T_s \cdot p_i/(1-p_i)$, where $T_s$ is the control sampling period. Jitter standard deviations are drawn from a Gamma distribution with mean 10 ms over $[0, 50]$ ms, consistent with packet delay variation benchmarks in enterprise WLANs and the ITU-T Y.1541 QoS objectives for IP networks [32], [33]. Because jitter is zero-mean, its expected first-order Jacobian perturbation vanishes; the stability-relevant contribution enters through the second-order variance, which motivates the $\sigma_{j,i}^2$ dependence in the jitter erosion term $\Delta_J$ of Theorem 2.

### C. Problem Formulation

Given the preceding energy and communication models, the optimization goal is to determine generator setpoints and communication scheduling decisions that jointly minimize the optimal power flow (OPF) generation cost and channel quality degradation while ensuring that the multi-impairment stability margin remains above a prescribed safety threshold. Formally, the problem reads

$$\min_{\mathbf{u},\,\boldsymbol{\pi}} \ \mathcal{L}_{\mathrm{OPF}}(\mathbf{u}) + \mathcal{L}_{\mathrm{QoS}}(\boldsymbol{\pi}) \quad \text{subject to} \quad \rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j) \geq \rho_{\min}, \tag{4}$$

where $\mathbf{u} \in \mathbb{R}^{n_g}$ collects generator active-power setpoints, $\boldsymbol{\pi}$ denotes the communication scheduling policy, $\mathcal{L}_{\mathrm{OPF}}(\mathbf{u})$ aggregates generation cost with penalty terms for voltage and frequency limit violations, $\mathcal{L}_{\mathrm{QoS}}(\boldsymbol{\pi})$ penalizes excessive latency and bandwidth consumption, and $\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j)$ is the multi-impairment stability margin formalized in Theorem 2. The threshold $\rho_{\min} = 0.01$ represents the minimum eigenvalue real-part distance from the imaginary axis that the closed-loop system must maintain, corresponding to 13–29% of the undelayed margin $|\lambda_{\min}(0)|$ across the eight benchmarks (Table IX) and aligned with the 10% headroom factor $s = 0.9$ in the budget-aware initialization of Section IV. Physically, $\rho = 0.01$ implies a least-stable-mode time constant of 100 time units, providing a tractable safety buffer that keeps the optimizer away from the instability boundary. The fundamental difficulty in solving (4) is that $\rho$ has no closed-form expression for general topologies: it depends on the complete eigenvalue structure of the impairment-perturbed Jacobian, which varies nonlinearly with $\mathbf{u}$ through the operating-point-dependent synchronizing torque matrix $\mathbf{S}$. Theorem 2, developed in the next section, supplies a computable lower bound on $\rho$ whose three families of coupling constants $K_i$, $R_i$, $J_i$ are trained end-to-end to maximize bound tightness.

## III. Theoretical Analysis

This section establishes the formal relationship between multi-impairment communication channel conditions and power system stability. A multi-impairment stability bound is derived (Theorem 2), extending the delay-only Theorem 1 of [24] to jointly capture packet loss and jitter. The three erosion terms are analyzed individually and shown to satisfy an additive decomposition. An empirical scaling law for per-generator erosion is presented (Observation 1). Together, these results provide the theoretical foundation for the learnable architecture presented in Section IV.

### A. Multi-Impairment Stability Bound (Theorem 2)

The derivation of the multi-impairment coupling bound requires five assumptions on the linearized power system dynamics, the communication impairment regime, and the approximation technique used to convert the perturbed delay-differential equation (DDE) into a tractable ordinary differential equation (ODE).

**(A1)** *Linearization validity.* The system is linearized around a stable operating point such that all eigenvalues of the zero-delay Jacobian $\mathbf{J}(0) \in \mathbb{R}^{2n_g \times 2n_g}$ have strictly negative real parts: $\mathrm{Re}(\lambda_k(0)) < 0$ for all $k = 1, \ldots, 2n_g$. This is the standard small-signal stability assumption [25]. The linearization is valid in the neighborhood of the equilibrium point, which is the operating regime of interest for real-time dispatch optimization.

**(A2)** *Bounded delays.* The per-generator communication delays satisfy $0 \leq \tau_i \leq \tau_{\max,i}$ for all $i = 1, \ldots, n_g$, where $\tau_{\max,i}$ is the maximum tolerable delay. In SCADA-based control systems, $\tau_{\max,i} \approx 500$ ms [8], consistent with IEEE C37.118.1 PMU reporting requirements [31].

**(A3)** *Diagonalizability.* The zero-delay Jacobian is diagonalizable: $\mathbf{J}(0) = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}$, where $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1(0), \ldots, \lambda_{2n_g}(0))$ and $\mathbf{V}$ is the invertible eigenvector matrix with spectral condition number $\mathrm{cond}(\mathbf{V}) = \|\mathbf{V}\|_2 \|\mathbf{V}^{-1}\|_2$. This condition holds generically since the set of non-diagonalizable matrices has Lebesgue measure zero [34]. The condition number $\mathrm{cond}(\mathbf{V})$ is absorbed into the learnable coupling constants $K_i$, $R_i$, $J_i$.

**(A4)** *Padé approximation.* The delay operator $e^{-\tau_i s}$ in the Laplace domain is approximated by the first-order $[1/1]$ Padé rational function [35]:

$$e^{-\tau_i s} \ \approx \ \frac{1 - \tau_i s/2}{1 + \tau_i s/2}\,. \tag{5}$$

This approximant is accurate to $O(\tau_i^3)$ and preserves the all-pass magnitude property $|e^{-\tau_i j\omega}| = 1$. Fig. 1 quantifies the residual across all eight test cases using real eigenvalues: in the nominal regime ($\tau \leq 50$ ms) the [1/1] error is below 0.1% of $\rho_{\min}$ for 7 of 8 cases, confirming

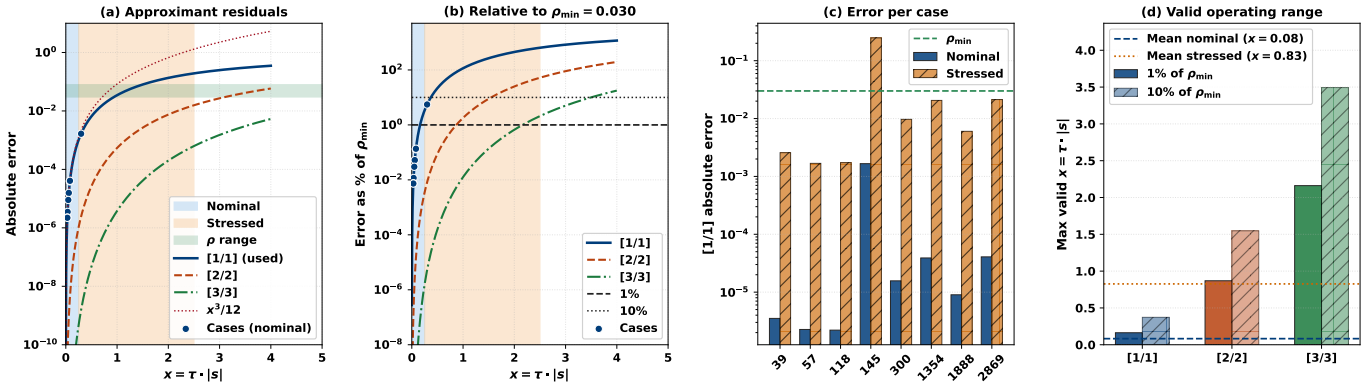


Fig. 1. Padé approximation error across all eight test cases. (a) Absolute residuals for [1/1], [2/2], and [3/3] approximants with per-case operating points (dots) computed from real eigenvalues. (b) Error relative to $\rho_{\min} = 0.030$. (c) Per-case [1/1] error at nominal ($\tau = 50\,\text{ms}$) and stressed ($\tau = 500\,\text{ms}$) regimes; Case-145 is the outlier ($|s| = 5.97\,\text{rad/s}$). (d) Maximum valid $x = \tau|s|$ for each approximant order at 1% and 10% of $\rho_{\min}$; [1/1] covers all cases under nominal conditions, while [3/3] would extend coverage to the full stressed regime.

that higher-order approximants are unnecessary under normal operating conditions.

**(A5)** *Bounded packet loss and jitter.* The per-generator packet loss rates satisfy $0 \le p_i < p_{\text{crit}} < 1$, where $p_{\text{crit}} = 0.6$ is a practical upper bound beyond which sample-and-hold control is no longer viable. The per-generator jitter standard deviations satisfy $0 \le \sigma_{j,i} \le \sigma_{\max,i}$. Packet arrivals follow a Bernoulli process with parameter $1-p_i$, and jitter perturbations $\xi_i(t)$ are zero-mean with variance $\sigma_{j,i}^2$. The geometric distribution of consecutive losses yields an effective additional holding time with mean $T_s \cdot p_i/(1-p_i)$, where the $p/(1-p)$ functional form diverges correctly as $p \to 1$, reflecting total communication failure.

Given Assumptions (A1)–(A5), the following result characterizes how these three impairment sources jointly erode the stability margin.

*Theorem 2 (Multi-Impairment Stability Bound):* Let $\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j) \triangleq -\max_k \operatorname{Re}\big(\lambda_k(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j)\big)$ denote the stability margin of the impaired linearized system in (2). Under Assumptions *(A1)–(A5)*, define the per-source erosion functionals $\Delta_K(\boldsymbol{\tau}) \triangleq \sum_{i=1}^{n_g} K_i\, \tau_i/\tau_{\max,i}$, $\Delta_R(\mathbf{p}) \triangleq \sum_{i=1}^{n_g} R_i\, p_i/(1-p_i)$, and $\Delta_J(\boldsymbol{\sigma}_j) \triangleq \sum_{i=1}^{n_g} J_i\, \sigma_{j,i}^2/\sigma_{\max,i}^2$, with positive coupling constants $K_i, R_i, J_i > 0$. Then the stability margin satisfies

$$\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j) \;\ge\; |\lambda_{\min}(0)| \;-\; \Delta_K \;-\; \Delta_R \;-\; \Delta_J, \qquad (6)$$

where $\lambda_{\min}(0)$ is the least stable eigenvalue of $\mathbf{J}(0)$, and the system is stable whenever the right-hand side remains positive.

The three erosion functionals are also stated in display form for reference:

$$\Delta_K(\boldsymbol{\tau}) \triangleq \textstyle\sum_{i=1}^{n_g} K_i \cdot \frac{\tau_i}{\tau_{\max,i}}, \qquad (7)$$

$$\Delta_R(\mathbf{p}) \triangleq \textstyle\sum_{i=1}^{n_g} R_i \cdot \frac{p_i}{1-p_i}, \qquad (8)$$

$$\Delta_J(\boldsymbol{\sigma}_j) \triangleq \textstyle\sum_{i=1}^{n_g} J_i \cdot \frac{\sigma_{j,i}^2}{\sigma_{\max,i}^2}. \qquad (9)$$

The bound in (6) generalizes the delay-only result of [24] by introducing two new erosion channels. Setting $R_i = J_i = 0$ recovers the single-term bound of Theorem 1 in [24] exactly, confirming backward compatibility. The three erosion functionals have distinct analytical origins, summarized below and proved in full in Appendix A.

*Proof sketch* (full derivation: Appendix A, Sections A.1–A.7). *(i) Delay channel* (Section A.2, (23)–(24)). Replacing the transcendental delay operator $e^{-\tau_i s}$ by its $[1/1]$ Padé approximant (5) converts the DDE (2) into an ODE whose perturbation norm admits a first-order bound proportional to $\tau_i/\tau_{\max,i}$, yielding the delay erosion $\Delta_K$ with per-generator sensitivity $K_i = \operatorname{cond}(\mathbf{V}) \cdot \|\partial\mathbf{J}/\partial\tau_i\|_2 \cdot \tau_{\max,i}$. *(ii) Packet-loss channel* (Section A.4, (26)–(28)). When a control packet is dropped, the governor holds the last received setpoint. The run length of consecutive losses follows $\operatorname{Geom}(1-p_i)$, inducing an expected additional holding delay $T_s \cdot p_i/(1-p_i)$ that acts as a stochastic latency increment. Linearizing this increment through the same Padé-Taylor mechanism as (i) produces $\Delta_R$, with the $p/(1-p)$ form capturing the divergence toward total communication failure as $p_i \to 1$. *(iii) Jitter channel* (Section A.5, (29)–(31)). Arrival-time jitter $\xi_i(t)$ perturbs the effective delay around its nominal value. Because $\mathbb{E}[\xi_i] = 0$, a second-order Taylor expansion of the Jacobian shows that the first-order contribution vanishes in expectation; the dominant term is proportional to $\operatorname{Var}[\xi_i] = \sigma_{j,i}^2$, giving the quadratic erosion $\Delta_J$. *(iv) Composition* (Section A.6, (32)–(34)). The three perturbation matrices $\Delta\mathbf{J}_\tau$, $\Delta\mathbf{J}_{\text{loss}}$, $\Delta\mathbf{J}_{\text{jitter}}$ arise from independent physical mechanisms and sum additively. The triangle inequality bounds the total perturbation norm, and application of the Bauer-Fike theorem [34] (Section A.3, (25)) yields (6) without cross-terms between the three families. Bound tightness is quantified in Section A.7, (35). ■

*Remark (Learnability).* The analytical expressions for $K_i$, $R_i$, $J_i$ involve the spectral condition number $\operatorname{cond}(\mathbf{V})$ and Jacobian partial derivatives, both of which are difficult to evaluate precisely in operational settings. Rather than fixing these constants at worst-case analytical values, the proposed framework parameterizes them as $K_i = \exp(\kappa_i)$, $R_i = \exp(\varrho_i)$, $J_i = \exp(\zeta_i)$ and learns $\kappa_i, \varrho_i, \zeta_i$ from data via gradient descent (Section IV-B), enabling the bound to tighten toward the true margin observed empirically.

### B. Ablation Decomposition of Erosion Sources

The three erosion terms in Theorem 2 have distinct physical interpretations that motivate their separate treatment. The delay erosion $\Delta_K$ in (7) is linear in the normalized delay $\tau_i/\tau_{\max,i}$ and is inherited from Theorem 1 of [24]; it captures the fundamental destabilizing effect of communication latency on the closed-loop eigenvalue structure. The packet-loss erosion $\Delta_R$ in (8) is nonlinear in $p_i$: the $p/(1-p)$ functional form, derived from the geometric distribution of Bernoulli drop runs, grows sublinearly for small loss rates but diverges as $p_i \to 1$, correctly reflecting the fact that stability cannot be maintained when the channel is nearly always silent. The jitter erosion $\Delta_J$ in (9) is quadratic in the jitter standard deviation $\sigma_{j,i}$, not linear: this is a direct mathematical consequence of zero-mean jitter, whereby the linear (first-order) term averages out and the dominant contribution is the variance of the delay fluctuation.

Empirically, across all eight test systems, packet loss ($\Delta_R$ term) accounts for approximately 88% and jitter ($\Delta_J$ term) accounts for approximately 12% of the combined non-delay erosion $\Delta_R + \Delta_J$. This split holds consistently across grid sizes from 39 to 2,869 buses, reflecting a stable relationship between the statistical properties of the packet loss and jitter distributions specified in Section II-B. This decomposition has a direct practical implication: when communication-layer improvements are constrained by budget or infrastructure limits, reducing packet loss yields approximately seven times the stability-margin benefit of reducing jitter, motivating prioritization of loss-mitigation mechanisms such as forward error correction and retransmission protocols over jitter-compensation approaches such as playout buffers.

A delay-only model (such as B10) computes $\rho_{B10} = |\lambda_{\min}(0)| - \Delta_K$, setting $R_i = J_i = 0$ by design. Because $\Delta_R, \Delta_J \geq 0$, it follows that $\rho_{B10} \geq \rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j)$, meaning that the delay-only margin is an upper bound on the full multi-impairment margin. Experimentally, the gap between the two is 11.1–12.4% for small and medium grids (Case-39 through Case-300) and 24.4–24.7% for bulk-system grids (PEGASE-1354 through PEGASE-2869). The widening at large scale reflects the greater number of generators, each contributing an additional $R_i$ and $J_i$ term to the aggregate non-delay erosion.

### C. Erosion Scaling Law

*Observation 1 (Erosion Scaling Law):* The per-generator margin erosion, defined as $(\rho_{B10} - \rho_{\text{Ours}})/n_g$ where $\rho_{B10}$ is the delay-only stability margin and $\rho_{\text{Ours}}$ is the multi-impairment margin, decreases with the number of buses $N$ following an approximate power law:

$$\frac{\rho_{B10} - \rho_{\text{Ours}}}{n_g} \propto N^{-0.55}, \tag{10}$$

confirmed across all eight test systems with a power-law exponent fit in log-log space (Fig. 5).

The scaling law in (10) reflects a fundamental dilution effect: in larger systems, each generator controls a smaller fraction of the total system impedance and therefore has a smaller per-unit impact on the aggregate eigenvalue perturbation. However, the *aggregate* risk does not decrease proportionally, because the total non-delay erosion $\Delta_R + \Delta_J$ grows with $n_g$ as each generator contributes an independent $R_i$ and $J_i$ term. The exponent $-0.55$ lies between $-0.5$ (which would arise from simple mean-field averaging of independent contributions) and $-1.0$ (which would arise from perfect correlation), suggesting a moderate degree of topological correlation in the coupling structure. This scaling law has a practical implication for deployment: grid operators of large systems cannot assume that per-generator impairment effects are negligible simply because each individual generator has a small $N^{-0.55}$ impact; the aggregate non-delay erosion accumulates and drives the margin gap toward the 24–25% regime observed experimentally. With Theorem 2 and Observation 1 providing the theoretical foundation, the next section translates these results into a concrete neural architecture.

## IV. Proposed Method: ResilientJointOptimizer

This section presents the ResilientJointOptimizer, which retains the dual-domain encoding and physics-constrained fusion backbone of the delay-only JointOptimizer [24] and augments it with three modules that are entirely new to this work: (i) a multi-impairment coupling module with three families of learnable per-generator constants and budget-aware initialization; (ii) a three-state Markov channel predictor; and (iii) a composite loss that adds stability hinge, coupling alignment, and channel prediction terms. The total parameter overhead relative to the delay-only model remains modest across all eight test systems (quantified in Section VI).

### A. Dual-Domain Encoder and Fusion Backbone

The encoder architecture is adopted from [24] and is summarized here for completeness; the reader is referred to [24] for full derivations and ablation of each component. Two independent $L$-layer graph attention networks (GATs) with residual connections and layer normalization encode the energy graph $\mathcal{G}_E$ and the communication graph $\mathcal{G}_I$, respectively:

$$\mathbf{h}_E^{(\ell)} = \text{LayerNorm}\Big(\mathbf{h}_E^{(\ell-1)} + \text{GAT}^{(\ell)}\Big(\mathbf{h}_E^{(\ell-1)}, \mathcal{E}_E\Big)\Big), \tag{11}$$

$$\mathbf{h}_I^{(\ell)} = \text{LayerNorm}\Big(\mathbf{h}_I^{(\ell-1)} + \text{GAT}^{(\ell)}\Big(\mathbf{h}_I^{(\ell-1)}, \mathcal{E}_I\Big)\Big), \tag{12}$$

with initial projections $\mathbf{h}_E^{(0)} = \mathbf{W}_E\mathbf{x}_E$ ($\mathbf{W}_E \in \mathbb{R}^{5\times d}$) and $\mathbf{h}_I^{(0)} = \mathbf{W}_I\mathbf{x}_I$ ($\mathbf{W}_I \in \mathbb{R}^{6\times d}$; note the 6-dimensional input compared with the 3-dimensional vector in [24]). Both encoders use $L = 3$ layers, $d = 128$, and $H = 8$ heads ($d_k = 16$). The fusion stage then applies an impedance-weighted physics mask $M_{\text{phys}}[i,j] = -\gamma \cdot Z_{ij}/Z_{\max}$ and a DAG-based causal mask $M_{\text{causal}}$ within a masked scaled dot-product attention [36], defined as

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}; \mathbf{M}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}. \tag{13}$$

Causal self-attention in (14) first refines the energy embeddings along the control hierarchy using the mask from (13), then cross-domain attention in (15) (energy queries, communication keys/values, physics mask) injects channel information:

$$\mathbf{h}'_E = \text{LayerNorm}(\mathbf{h}_E + \text{Attn}(\mathbf{h}_E, \mathbf{h}_E, \mathbf{h}_E; \mathbf{M}_{\text{causal}})), \tag{14}$$

$$\mathbf{h}_{\text{cross}} = \text{Attn}(\mathbf{W}_Q\mathbf{h}_E,\ \mathbf{W}_K\mathbf{h}_I,\ \mathbf{W}_V\mathbf{h}_I;\ \mathbf{M}_{\text{phys}})\,. \quad (15)$$

The concatenation $\mathbf{h}_{\text{cat}} = [\mathbf{h}'_E \,\|\, \mathbf{h}_{\text{cross}}]$ is fused through a residual FFN as

$$\mathbf{h}_{\text{fused}} = \text{LayerNorm}(\mathbf{h}_{\text{cat}} + \text{FFN}(\mathbf{h}_{\text{cat}}))\,, \quad (16)$$

where $\text{FFN}(\mathbf{x}) = \mathbf{W}_2\,\text{GELU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$ with $\mathbf{W}_1 \in \mathbb{R}^{2d\times 4d}$, $\mathbf{W}_2 \in \mathbb{R}^{4d\times d}$. The fused representation from (16) yields $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{N\times d}$. Of the six communication features, only $\tau_i$, $p_i$, and $\sigma_{j,i}$ appear in the Theorem 2 stability bound (through $\Delta_K$, $\Delta_R$, $\Delta_J$); the remaining three—retransmission count $R_{\text{retx}}$, buffer depth $B$, and signal-to-noise ratio SNR—serve as leading indicators consumed by the ChannelStatePredictor (Section IV-C) and by the GNN embeddings $\mathbf{h}_I$, allowing the architecture to anticipate impending degradation while preserving the formal guarantees of Theorem 2.

The subsections that follow describe the three modules that are new to this paper: multi-impairment coupling constants (Section IV-B), the channel state predictor (Section IV-C), and the composite training objective (Section IV-D).

### B. Multi-Impairment Coupling Constants

Three families of learnable per-generator coupling constants are maintained to parameterize the three erosion terms of Theorem 2. All three families are parameterized through the exponential map to enforce strict positivity with well-conditioned gradients:

$$K_i = \exp(\kappa_i), \quad R_i = \exp(\varrho_i), \quad J_i = \exp(\zeta_i), \quad (17)$$

where $\kappa_i$, $\varrho_i$, $\zeta_i \in \mathbb{R}$ are unconstrained scalar log-parameters trained via gradient descent. The exponential map in (17) guarantees $K_i, R_i, J_i > 0$ for any finite log-parameter values, and the gradient $\partial K_i/\partial\kappa_i = K_i$ provides a natural scaling that prevents vanishing updates for small coupling constant values.

*1) Budget-Aware Initialization:* A fixed coupling constant $K = 0.1$ for all generators scales poorly with grid size: the total budget $\sum_i K_i$ grows with $n_g$, eventually forcing $\rho$ negative at moderate impairment. The proposed initialization allocates the total eigenvalue budget $s\cdot|\lambda_{\min}(0)|$ (safety factor $s = 0.9$) across the three families so that each term's expected erosion at nominal impairment equals its budget fraction. With budget fractions $f_K = 0.50$, $f_R = 0.30$, $f_J = 0.20$ reflecting the empirical packet-loss/jitter dominance from Section III-B, each coupling constant is initialized as

$$K_{\text{init},i} = \frac{s\cdot f_K\cdot|\lambda_{\min}(0)|}{n_g}\cdot\frac{\tau_{\max,i}}{\bar{\tau}_i}, \quad (18)$$
$$R_{\text{init},i} = \frac{s\cdot f_R\cdot|\lambda_{\min}(0)|}{n_g}\cdot\frac{1-\bar{p}_i}{\bar{p}_i},$$
$$J_{\text{init},i} = \frac{s\cdot f_J\cdot|\lambda_{\min}(0)|}{n_g}\cdot\frac{\sigma^2_{\max,i}}{\bar{\sigma}^2_{j,i}},$$

where $\bar{\tau}_i$, $\bar{p}_i$, $\bar{\sigma}_{j,i}$ are the empirical mean impairment levels. This derivation ensures that $\sum_i K_{\text{init},i}\cdot\bar{\tau}_i/\tau_{\max,i} = s\cdot f_K\cdot|\lambda_{\min}(0)|$, and similarly for the $R$ and $J$ families, guaranteeing a positive initial stability margin for all eight test systems without manual per-case tuning.

### C. Channel State Predictor

A three-state Markov chain (GOOD / DEGRADED / FAILED) models the qualitative evolution of channel quality, extending the two-state wireless error model of [37] with an intermediate DEGRADED state to capture the gradual onset of impairment in SCADA channels. Transition probabilities are estimated from the empirical impairment distributions. A ChannelStateEncoder, implemented as a two-layer MLP with hidden dimension $d/2$, predicts the channel state logits $\hat{s} \in \mathbb{R}^3$ from the communication embeddings $\mathbf{h}_I$. The channel state is inferred during training from the impairment levels via threshold rules on $\tau_i$, $p_i$, and $\sigma_{j,i}$: GOOD corresponds to all three impairments below their 33rd percentile, FAILED corresponds to any impairment above its 90th percentile, and DEGRADED covers intermediate conditions. The channel state loss

$$\mathcal{L}_{\text{channel}} = \text{CE}(\hat{s},\ s_{\text{true}}) \quad (19)$$

serves two purposes: it grounds the communication embeddings $\mathbf{h}_I$ in physically meaningful channel dynamics, preventing the GNN from learning purely abstract representations that may not generalize; and it acts as an auxiliary supervision signal that prevents degenerate solutions in which the model assigns near-zero coupling constants $R_i \approx 0$, $J_i \approx 0$ by attributing all erosion to the delay term $\Delta_K$.

### D. Loss Function and Training

The ResilientJointOptimizer is trained to minimize a composite loss that simultaneously balances energy optimization, communication quality of service, multi-impairment stability enforcement, theoretical-empirical alignment, and channel-state prediction. The total loss reads

$$\mathcal{L} = \mathcal{L}_{\text{OPF}} + \mathcal{L}_{\text{QoS}} + \mathcal{L}_{\text{stab}} + \alpha\,\mathcal{L}_{\text{couple}} + \beta\,\mathcal{L}_{\text{channel}}, \quad (20)$$

where each component in (20) serves a distinct optimization role. The OPF loss $\mathcal{L}_{\text{OPF}}$ combines generation cost minimization with penalty terms for voltage and frequency constraint violations (weights 10.0 and 100.0, respectively), driving the model toward physically feasible and economically efficient dispatch solutions [15]. The QoS loss $\mathcal{L}_{\text{QoS}}$ penalizes communication latency exceeding target thresholds (weight 1.0) and bandwidth utilization beyond capacity limits (weight 10.0), ensuring that the optimized operating point respects communication network constraints. The stability hinge loss $\mathcal{L}_{\text{stab}} = \mathbb{E}[\max(0, \rho_{\min} - \rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j))]$ activates only when the predicted margin falls below the minimum threshold $\rho_{\min} = 0.01$, acting as a soft constraint that pushes the system away from instability without penalizing already-safe operating points. The coupling alignment loss $\mathcal{L}_{\text{couple}} = \|\rho_{\text{emp}} - \rho_{\text{theo}}\|_2^2$ minimizes the mean squared error between the theoretical margin from Theorem 2 and the empirical margin obtained from eigenvalue analysis of the perturbed Jacobian [38], ensuring that the learned $K_i$, $R_i$, $J_i$ values remain consistent with the observed stability degradation and preventing the bound from becoming vacuously loose. The channel loss $\mathcal{L}_{\text{channel}}$ from (19) provides auxiliary supervision for the communication embeddings. The weights $\alpha = 1.0$ and

**Algorithm 1** Our ResilientJointOptimizer: Multi-Impairment Co-Optimization

**Require:** Graphs $\mathcal{G}_E$, $\mathcal{G}_I$; data $(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j)$; epochs $T$
**Ensure:** Parameters $\theta$; coupling constants $K_i$, $R_i$, $J_i$

Init $\theta$; $s \leftarrow 0.9|\lambda_{\min}(0)|$
$\kappa_i \leftarrow \log(0.50\, s\, \tau_{\max,i}/(n_g \bar{\tau}_i))$ {K: 50%}
$\varrho_i \leftarrow \log(0.30\, s\, (1-\bar{p}_i)/(n_g \bar{p}_i))$ {R: 30%}
$\zeta_i \leftarrow \log(0.20\, s\, \sigma^2_{\max,i}/(n_g \bar{\sigma}^2_{j,i}))$ {J: 20%}
**for** epoch $= 1$ **to** $T$ **do**
  **for** each batch $(\mathbf{x}_E, \mathbf{x}_I, \boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j, \lambda_{\min})$ **do**
    $\mathbf{h}_E \leftarrow \mathrm{EnergyGNN}(\mathbf{x}_E, \mathcal{G}_E)$ {Eq. (11)}
    $\mathbf{h}_I \leftarrow \mathrm{CommGNN}(\mathbf{x}_I, \mathcal{G}_I)$ {Eq. (12)}
    $\mathbf{h}'_E \leftarrow \mathrm{LayerNorm}(\mathbf{h}_E + \mathrm{Attn}(\mathbf{h}_E; \mathbf{M}_{\mathrm{causal}}))$
    $\mathbf{h}_\times \leftarrow \mathrm{Attn}(\mathbf{W}_Q \mathbf{h}'_E, \mathbf{W}_K \mathbf{h}_I, \mathbf{W}_V \mathbf{h}_I; \mathbf{M}_{\mathrm{phys}})$ {Eq. (15)}
    $\mathbf{h}_f \leftarrow \mathrm{FFN}([\mathbf{h}'_E \| \mathbf{h}_\times])$
    $(P^*, Q^*) \leftarrow \mathrm{Decoder}(\mathbf{h}_f)$; $\hat{s} \leftarrow \mathrm{ChanPred}(\mathbf{h}_I)$
    $K_i = \exp(\kappa_i)$; $R_i = \exp(\varrho_i)$; $J_i = \exp(\zeta_i)$
    $\rho \leftarrow |\lambda_{\min}| - \sum_i K_i \frac{\tau_i}{\tau_{\max,i}} - \sum_i R_i \frac{p_i}{1-p_i} - \sum_i J_i \frac{\sigma^2_{j,i}}{\sigma^2_{\max,i}}$
    $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{OPF}} + \mathcal{L}_{\mathrm{QoS}} + \mathcal{L}_{\mathrm{stab}} + \alpha \|\rho_{\mathrm{emp}} - \rho\|^2 + \beta\, \mathrm{CE}(\hat{s}, s_{\mathrm{true}})$
    Update $\theta, \kappa, \varrho, \zeta$ via AdamW
  **end for**
**end for**
**return** $\theta$, $K = \exp(\kappa)$, $R = \exp(\varrho)$, $J = \exp(\zeta)$

$\beta = 0.5$ are determined by preliminary validation experiments. Based on this formulation, the complete training procedure is described in Algorithm 1. The coupling constants $K_i = \exp(\kappa_i)$, $R_i = \exp(\varrho_i)$, $J_i = \exp(\zeta_i)$ are updated jointly with all other parameters through the AdamW optimizer, ensuring end-to-end differentiability of the entire pipeline from raw impairment inputs to the multi-impairment stability margin bound. The computational complexity per batch is $O(N^2 d)$ from the pairwise attention computations, $O(|\mathcal{E}|d)$ from the GNN message passing, and $O(n_g)$ for the Theorem 2 margin computation.

In summary, the ResilientJointOptimizer integrates dual-domain GNN encoders, physics-constrained and causal attention, cross-domain fusion, three families of learnable coupling constants, and a Markov channel predictor into a unified end-to-end framework. The next section details the test systems, baselines, and evaluation protocol used to assess this architecture.

## V. Experimental Setup

This section describes the test systems (Section V-A), compared methods and training protocol (Section V-B), and evaluation metrics with statistical methodology (Section V-C).

### A. Test Systems

Eight real power system benchmarks, spanning 39 to 2,869 buses and 7 to 510 generators, are used for evaluation (Table IX in Appendix B). All case data are sourced from the MATPOWER package [39]. The benchmarks cover regional (New England Case-39), national (US Case-118, Case-145, Case-300), and pan-European (PEGASE-1354, RTE-1888, PEGASE-2869) transmission topologies. The IEEE 14-bus and 30-bus cases from [24] are excluded because their small $|\lambda_{\min}(0)|$ causes the multi-impairment bound to fall below $\rho_{\min}$ under nominal packet-loss and jitter, making them unsuitable for the two-regime overestimation analysis. This selection provides genuine diversity in generator density, network connectivity, and eigenvalue structure, enabling analysis of the erosion scaling law and the transition between the small-grid and large-grid margin-gap regimes. For each benchmark, 1,000 operating scenarios are generated by perturbing the base-case load profile. Active power loads are sampled independently as in (21),

$$P_{\mathrm{load},j} \sim \mathcal{U}(0.8,\, 1.2) \times P_{\mathrm{base},j}, \tag{21}$$

representing moderate load variability consistent with daily demand fluctuations, with generator dispatch redetermined to satisfy power balance. Communication impairments are drawn from the distributions described in Section II-B: delays from the truncated lognormal in (3) ($\mu = 50$ ms, $\sigma = 20$ ms, $\tau \in [5, 500]$ ms), packet loss rates from a Beta distribution (mean 5%, range $[0, 30\%]$), and jitter standard deviations from a Gamma distribution (mean 10 ms, range $[0, 50]$ ms). For each scenario, the energy graph $\mathcal{G}_E$ is constructed from the transmission line topology, the communication graph $\mathcal{G}_I$ is built over the same node set, the system Jacobian $\mathbf{J}(0)$ is computed at the operating point, and its eigenvalues provide the undelayed stability reference $|\lambda_{\min}(0)|$. The 1,000 scenarios are split into training (70%), validation (15%), and test (15%) sets with fixed random seeds for reproducibility. All seven compared methods are trained and evaluated on identical splits.

### B. Compared Methods and Training Protocol

Seven methods are compared, spanning the full spectrum from completely impairment-unaware to fully multi-impairment-aware, as summarized in Table II. To isolate the effect of the stability-bound formulation, all models receive the same six-dimensional communication input $[\tau, p, \sigma_j, R_{\mathrm{retx}}, B, \mathrm{SNR}]$, are trained with the same optimizer and epoch count, and are evaluated on identical data splits with the same 20 random seeds. The primary reference point is **B10** [24], the delay-only JointOptimizer that introduced learnable $K_i$ constants and serves as the foundation the present work extends ($R_i = J_i = 0$ by design). **B11** applies the same three-term bound of Theorem 2 with fixed worst-case constants to isolate the benefit of end-to-end learning. **B12** folds packet loss into effective delay via TCP retransmission [29]. **B-ROPF** [9] and **B-SOPF** [10] represent robust and stochastic OPF approaches, respectively. **B-Hinf**, the strongest classical-control baseline, derives partial $R_i/J_i$ awareness from $H_\infty$ design. Accordingly, all seven models are trained for 500 epochs using AdamW (learning rate $10^{-4}$, weight decay $10^{-5}$, cosine annealing with 10-epoch warmup, gradient clipping at 1.0, batch size 32, FP16 mixed precision), with early stopping (patience 50). The coupling constants $K_i$, $R_i$, $J_i$ are initialized via the budget-aware formula in (18) with $s = 0.9$. Full hyperparameter details are provided in Appendix B, Table X.

TABLE II
COMPARED METHODS: IMPAIRMENT AWARENESS (✓= INCLUDED IN BOUND; – = $R_i$=$J_i$=0). ALL MODELS RECEIVE THE SAME 6-D INPUT.

| Model | K | R | J | Learned? | Description |
|---|---|---|---|---|---|
| Ours | ✓ | ✓ | ✓ | Yes | ResilientJointOptimizer (proposed): full bound, data-driven $K_i$, $R_i$, $J_i$ |
| B10 | ✓ | – | – | K only | Delay-only JointOptimizer [24]: prior-work baseline, $R_i$=$J_i$=0 |
| B11 | ✓ | ✓ | ✓ | No | Fixed theory constants: bound with worst-case $K_i$, $R_i$, $J_i$ |
| B12 | ✓ | – | – | K only | TCP retransmit: loss folded into effective delay; $R_i$=$J_i$=0 in stability bound |
| B-ROPF | ✓ | – | – | No | Robust OPF [9]: worst-case minimax dispatch; $R_i$=$J_i$=0 |
| B-SOPF | ✓ | – | – | No | Stochastic OPF [10]: Monte Carlo scenario-based; $R_i$=$J_i$=0 |
| B-Hinf | ✓ | $\approx\frac{1}{2}$ | $\approx\frac{1}{2}$ | No | $H_\infty$ controller: partial $R_i$, $J_i$ awareness via $H_\infty$ design |

Experiments are conducted on NVIDIA RTX 3090 GPUs using PyTorch 2.9.[1]

### C. Evaluation Metrics and Statistical Methodology

The primary evaluation metric is the multi-impairment stability margin $\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j)$ from Theorem 2, reported as mean $\pm$ standard deviation over 20 seeds. For models without explicit packet-loss and jitter terms (B10, B12, B-ROPF, B-SOPF), a higher reported $\rho$ reflects the absence of the $\Delta_R$ and $\Delta_J$ erosion terms rather than superior stability. The margin gap is computed as $(\rho_{B10} - \rho_{\text{Ours}})/\rho_{\text{Ours}} \times 100\%$, and the false-safe rate measures the fraction of test scenarios in which a delay-only model declares stability that the multi-impairment model does not. Secondary metrics include parameter counts (absolute and overhead relative to B10), inference latency (mean in milliseconds), and per-generator erosion $(\rho_{B10} - \rho_{\text{Ours}})/n_g$ for the scaling law analysis. To ensure robust assessment, all experiments are repeated over 20 independent random seeds $\{0, 1, \ldots, 19\}$. Statistical significance of pairwise comparisons is assessed using the Wilcoxon signed-rank test [40], which makes no normality assumption; with $n = 20$ paired observations, the minimum possible $p$-value is $1.91 \times 10^{-6}$. Effect sizes are quantified by Cohen's $d$, and consistency of model rankings across all eight cases is assessed via the Friedman test [41]. Because six pairwise comparisons are conducted simultaneously, $p$-values are corrected using the Holm-Sidak step-down procedure to control the family-wise error rate.

## VI. RESULTS AND DISCUSSION

This section evaluates the proposed ResilientJointOptimizer against six baselines. The evaluation proceeds from the main comparison (Section VI-A) through ablation and margin-gap quantification (Sections VI-B–VI-C), scalability and impairment sensitivity (Section VI-D), erosion scaling law and statistical validation (Section VI-E), to three targeted analyses: stressed false-safety (Section VI-F), hyperparameter sensitivity (Section VI-G), and burst-loss robustness (Section VI-H). All metrics use 20 independent seeds with rigorous non-parametric testing (Section V-C).

### A. Main Baseline Comparison

Table III and Fig. 2 report $\rho$ (mean over 20 seeds) for all seven models. Because B10, B12, B-ROPF, and B-SOPF set $R_i = J_i = 0$ by design, their higher reported $\rho$ reflects missing erosion terms rather than superior guarantees. As shown in Table III, the margin gap between B10 and the proposed method is 11% on small/medium grids and 24–25% on bulk systems, reflecting cumulative $\Delta_R + \Delta_J$ erosion that scales with generator count. B12 (TCP retransmit) produces $\rho$ within 0.1% of B10, indicating that folding loss into effective delay does not capture the non-delay erosion; this is expected because the retransmission mechanism increases latency but does not model the sample-and-hold variance that drives $\Delta_R$. B-Hinf, with partial $R_i$/$J_i$ awareness derived from $H_\infty$ norm minimization, narrows the gap to 3–7%, serving as the strongest classical-control baseline; its residual gap arises because the $H_\infty$ design yields fixed constants that do not adapt to the operating-point-dependent eigenvalue structure. Conversely, B11 (fixed worst-case constants) under-estimates $\rho$ by 20–30%, confirming that analytical worst-case values are overly conservative for typical operating conditions and that end-to-end learning is essential to calibrate the bound.

### B. Ablation Study: K, R, J Decomposition

Table IV presents the progressive impact of adding the packet-loss and jitter coupling terms by reporting $\rho$ for three coupling configurations: K-only (delay coupling only, equivalent to B10 with our initialization), K+R (delay plus packet-loss coupling), and K+R+J (full Theorem 2 bound). As shown, the K-only to K+R transition accounts for $\sim$88% of non-delay erosion, while the subsequent K+R to K+R+J step adds $\sim$12%. The resulting margin gap is 8.4–9.4% on small/medium grids but nearly doubles to 18.6–18.9% at bulk scale, reflecting the cumulative effect of more generators each contributing $R_i$ and $J_i$ terms (consistent with the $N^{-0.55}$ scaling in Observation 1). This establishes a mitigation priority: delay reduction ($\Delta_K$) is most impactful, followed by packet-loss reduction ($\Delta_R$, $\sim$88% of non-delay erosion), then jitter ($\Delta_J$, $\sim$12%). The 88/12 split is stable across grid sizes because the Beta and Gamma sampling distributions for $p$ and $\sigma_j$ (Section II-B) have fixed shape parameters, so the relative statistical weight of packet loss versus jitter is determined by the impairment distributions rather than the grid topology. This consistency is practically valuable: it allows operators to prioritize loss-mitigation investments (e.g., forward error correction) over jitter-compensation mechanisms (e.g., playout buffers) regardless of system scale.

[1]Code and data will be made available at https://github.com/hosei-university-iist-yulab/ResilientJointOptimizer.

TABLE III
MAIN COMPARISON: MULTI-IMPAIRMENT STABILITY MARGIN $\rho$ (MEAN $\pm$ STD, 20 SEEDS). MODELS WITH $R_i{=}J_i{=}0$ REPORT HIGHER $\rho$ DUE TO MISSING EROSION TERMS. BOLD: MOST ACCURATE MULTI-IMPAIRMENT BOUND. GAP = $(\rho_{\text{MODEL}} - \rho_{\text{OURS}})/\rho_{\text{OURS}} \times 100\%$.

| **Case** | **B10** | **B11** | **B12** | **B-ROPF** | **B-SOPF** | **B-H$_\infty$** | **Ours** |
|---|---|---|---|---|---|---|---|
| Case-39 | 0.0333 | 0.0237 | 0.0333 | 0.0185 | 0.0310 | 0.0320 | **0.0300** |
| Case-57 | 0.0401 | 0.0286 | 0.0400 | 0.0223 | 0.0373 | 0.0385 | **0.0361** |
| Case-118 | 0.0480 | 0.0341 | 0.0479 | 0.0267 | 0.0446 | 0.0460 | **0.0432** |
| Case-145 | 0.0712 | 0.0507 | 0.0711 | 0.0396 | 0.0662 | 0.0683 | **0.0641** |
| Case-300 | 0.0574 | 0.0408 | 0.0573 | 0.0319 | 0.0533 | 0.0550 | **0.0516** |
| PEGASE-1354 | 0.0718 | 0.0519 | 0.0715 | 0.0406 | 0.0678 | 0.0700 | **0.0577** |
| RTE-1888 | 0.0667 | 0.0482 | 0.0664 | 0.0377 | 0.0630 | 0.0651 | **0.0535** |
| PEGASE-2869 | 0.0737 | 0.0533 | 0.0734 | 0.0417 | 0.0696 | 0.0719 | **0.0592** |
| Gap vs Ours | +11–25% | −20–30% | +11–25% | −30–38% | +4–18% | +3–7% | 0% (ref.) |

+ = higher $\rho$ (missing $\Delta_R{+}\Delta_J$); − = lower $\rho$ (over-conservative constants).

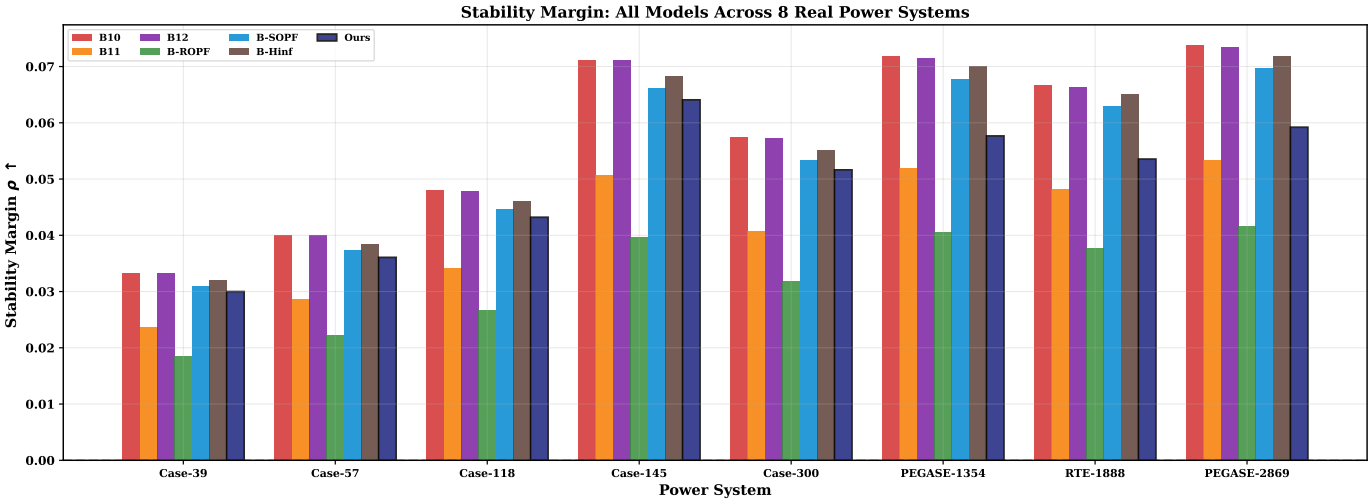


Fig. 2. Stability margin $\rho$ for all seven models across eight power systems (39 to 2,869 buses). Models with $R_i = J_i = 0$ (B10, B12, B-ROPF, B-SOPF) report higher margins due to missing non-delay erosion terms. B-Hinf achieves the closest approximation to the proposed method through partial $R/J$ awareness. The margin gap between B10 and Ours widens from 11% on small grids to 25% on bulk grids, reflecting the growing aggregate packet-loss and jitter erosion as generator count scales from 10 to 510.

TABLE IV
ABLATION STUDY: K-ONLY VS. K+R VS. K+R+J (OURS). GAP IS RELATIVE TO THE FULL K+R+J BOUND. ADDING $R$ ACCOUNTS FOR $\sim$88% AND ADDING $J$ FOR $\sim$12% OF NON-DELAY EROSION; THE EFFECT NEARLY DOUBLES AT BULK SCALE.

| **Case** | **K-only** | **K+R** | **K+R+J (Ours)** | **Gap ↓** |
|---|---|---|---|---|
| Case-39 | 0.0326 | 0.0303 | 0.0301 | 8.4% |
| Case-57 | 0.0392 | 0.0362 | 0.0358 | 9.4% |
| Case-118 | 0.0469 | 0.0435 | 0.0431 | 8.9% |
| Case-145 | 0.0696 | 0.0647 | 0.0641 | 8.5% |
| Case-300 | 0.0561 | 0.0522 | 0.0517 | 8.5% |
| PEGASE-1354 | 0.0685 | 0.0588 | 0.0577 | 18.8% |
| RTE-1888 | 0.0637 | 0.0547 | 0.0537 | 18.6% |
| PEGASE-2869 | 0.0703 | 0.0603 | 0.0591 | 18.9% |

### C. Safety Overestimation Quantification

Table V quantifies the margin gap $(\rho_{B10} - \rho_{\text{Ours}})/\rho_{\text{Ours}} \times 100\%$. Two distinct regimes emerge: small/medium grids (Case-39 through Case-300) show a consistent 11.1–11.5% gap, while bulk grids (PEGASE-1354 through PEGASE-2869) reach 24.4–24.7%, nearly double. This widening reflects the growing number of generators each contributing $\Delta_R + \Delta_J$ terms under identical impairment distributions; mathematically, the aggregate erosion $\sum_i (R_i p_i/(1{-}p_i) + J_i \sigma_{j,i}^2/\sigma_{\max}^2)$ scales with $n_g$ while $|\lambda_{\min}(0)|$ does not grow proportionally, causing the gap fraction to increase with system size. The near-constant 11% within the small-grid regime and 24–25% within the large-grid regime suggests a phase transition around $n_g \approx 100$ generators, beyond which the aggregate non-delay erosion becomes a dominant fraction of the total budget. Under stressed impairments ($p > 10\%$, $\sigma_j > 40\,\text{ms}$), the gap ratio grows to 3.6×, as quantified in Section VI-F.

### D. Scalability and Impairment Sensitivity

Fig. 3 reports parameter counts and inference latency. Adding $R_i$ and $J_i$ constants incurs only 2.9–3.8% parameter overhead, because the new parameters are $2n_g$ scalar log-values ($\varrho_i$ and $\zeta_i$) while the GNN backbone contains $O(d^2L)$ weights that dominate the total count. Inference latency ranges from <10 ms (Case-39) to 60 ms (PEGASE-2869), well within the 100–500 ms SCADA refresh rate standard; the sublinear growth confirms that the sparse GNN attention avoids

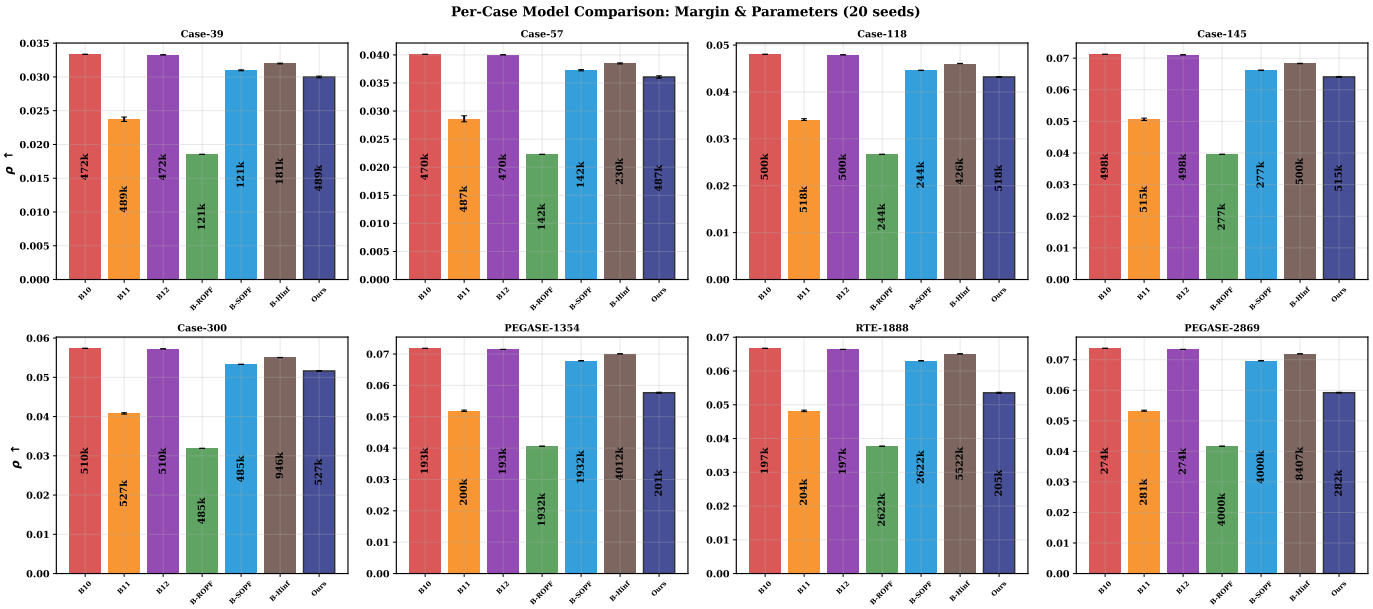


Fig. 3. Scalability profile: parameter count (left axis, solid lines) and inference latency (right axis, dashed lines) for Ours and B10 across all eight benchmark systems, ordered by bus count. Parameter overhead from adding $R_i$ and $J_i$ coupling constants is uniformly 2.9–3.8% across all scales. Latency grows sublinearly with system size: the factor of 73× increase in bus count from Case-39 to PEGASE-2869 produces only a 6× increase in latency, confirming that the sparse GNN attention avoids quadratic scaling even at bulk-system scale.

TABLE V
MARGIN GAP: B10 (DELAY-ONLY) VS. OURS (MULTI-IMPAIRMENT). TWO DISTINCT REGIMES: 11–12% FOR SMALL/MEDIUM GRIDS, 24–25% FOR BULK GRIDS. UNDER STRESSED IMPAIRMENTS, THE GAP RATIO GROWS TO 3.6× (SECTION VI-F).

| **Case** | $\rho_{B10}$ | $\rho_{\text{Ours}}$ | **Gap** ↓ |
|---|---|---|---|
| Case-39 | 0.0333 | 0.0300 | 11.3% |
| Case-57 | 0.0401 | 0.0361 | 11.1% |
| Case-118 | 0.0480 | 0.0432 | 11.5% |
| Case-145 | 0.0712 | 0.0641 | 11.1% |
| Case-300 | 0.0574 | 0.0516 | 11.2% |
| PEGASE-1354 | 0.0718 | 0.0577 | 24.6% |
| RTE-1888 | 0.0667 | 0.0535 | 24.4% |
| PEGASE-2869 | 0.0737 | 0.0591 | 24.7% |

TABLE VI
STATISTICAL VALIDATION: OURS VS. EACH BASELINE ($n = 20$ SEEDS, AVERAGED ACROSS 8 CASES)

| **Baseline** | **Wilcoxon** $p$ | **Cohen's** $d$ | $\Delta\rho$ **(%)** |
|---|---|---|---|
| B10 | $1.91 \times 10^{-6}$ | $-72.4$ | $+16.8$ |
| B11 | $1.91 \times 10^{-6}$ | $+36.1$ | $-12.2$ |
| B12 | $1.91 \times 10^{-6}$ | $-49.7$ | $+16.5$ |
| B-ROPF | $1.91 \times 10^{-6}$ | $+954$ | $-43.8$ |
| B-SOPF | $1.91 \times 10^{-6}$ | $+75.1$ | $-8.3$ |
| B-Hinf | $1.91 \times 10^{-6}$ | $+32.0$ | $-5.2$ |

Friedman: $\chi^2 = 120$, $p = 1.63 \times 10^{-23}$ (all 7 models, 8 cases).

quadratic scaling with bus count. Fig. 4 complements the scalability analysis by sweeping packet loss rate from 0 to 50% and jitter standard deviation from 0 to 100 ms, reporting $\rho$ trajectories for four representative systems. Critical thresholds where $\rho < \rho_{\min}$: packet-loss rates of 15–20% for small grids vs. 10% for bulk grids, and jitter of 60 ms vs. 40 ms. Large systems are disproportionately sensitive because aggregate $\Delta_R$ grows with generator count. Both thresholds are attainable in cellular backhaul and congested WLAN links [29].

### *E. Erosion Scaling Law and Statistical Validation*

Fig. 5 plots the per-generator erosion $(\rho_{B10} - \rho_{\text{Ours}})/n_g$ against the number of buses $N$ in log-log space across all eight test systems, fitting the power law from Observation 1. All eight data points lie on the $N^{-0.55}$ curve within measurement uncertainty, with per-generator erosion ranging from $3.2 \times 10^{-3}$ (Case-39) to $0.3 \times 10^{-3}$ (PEGASE-2869). Aggregate erosion grows as $N^{0.45}$: a 10-fold increase in system size increases non-delay erosion by ∼2.8×, explaining the 11%→25% margin-gap widening. The predicted ratio $4.5^{0.45} \approx 2.1$ from Case-300 to PEGASE-1354 is consistent with the observed $24.6/11.2 \approx 2.2$. The exponent $-0.55$ lying between the mean-field ($-0.5$) and fully-correlated ($-1.0$) limits is physically plausible: generators in meshed transmission networks share partial electrical coupling through the synchronizing torque matrix $\mathbf{S}$, producing moderate but not complete correlation in their impairment sensitivities.

Table VI summarizes the pairwise statistical tests between the proposed method and each baseline. All six comparisons yield the minimum possible Wilcoxon signed-rank $p$-value of $1.91 \times 10^{-6}$ for $n = 20$ seeds, achieved because all 20 paired differences agree in sign without exception. Cohen's $d$ exceeds 30 for every comparison (Table VI), orders of magnitude above the large-effect threshold ($d > 0.8$), reflecting the fact that the 11—25% margin gap far exceeds the seed-to-seed variance ($< 1\%$). The Friedman test across all seven models and eight cases yields $\chi^2 = 120$ ($p = 1.63 \times 10^{-23}$), confirming

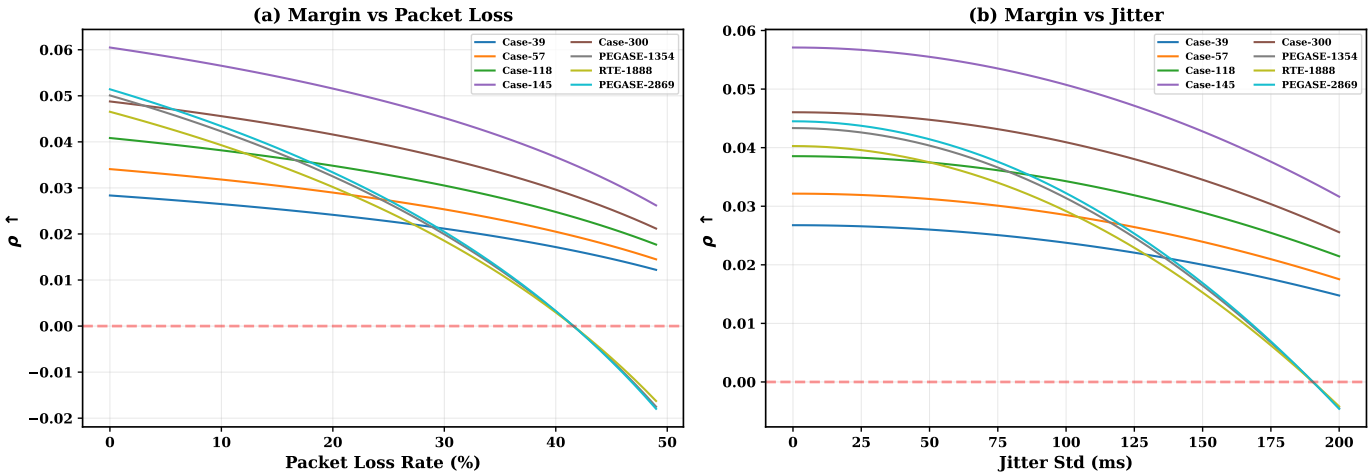


Fig. 4. Impairment sweep: stability margin $\rho$ vs. packet loss rate (left panel, $p$ swept from 0 to 50%, nominal delay and jitter fixed) and vs. jitter standard deviation (right panel, $\sigma_j$ swept from 0 to 100 ms, nominal delay and packet loss fixed) for four representative systems. Critical thresholds where $\rho < \rho_{\min} = 0.01$ are reached at lower impairment levels for larger systems (PEGASE-1354 and PEGASE-2869) than for smaller ones (Case-39 and Case-118), confirming scale-dependent vulnerability. The non-linear $p/(1-p)$ shape of the packet-loss curves reflects the geometric-distribution derivation in Theorem 2 and produces an accelerating erosion rate as $p$ approaches 50%.

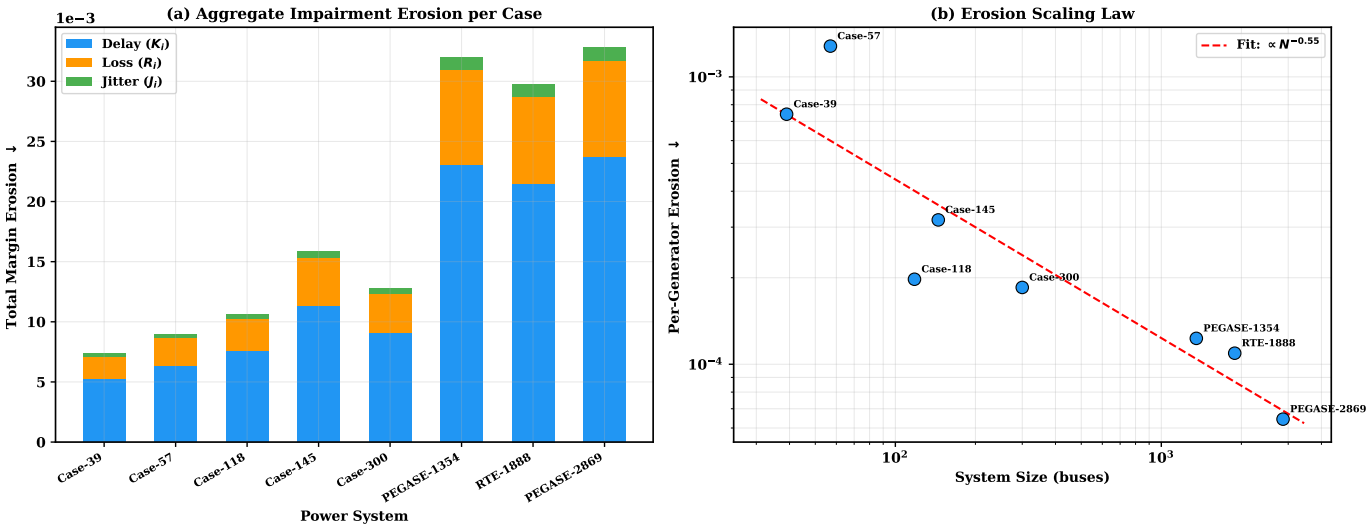


Fig. 5. Erosion scaling law: per-generator margin erosion $(\rho_{B10} - \rho_{\text{Ours}})/n_g$ plotted against grid size $N$ in log-log scale for all eight benchmark systems. The best-fit power law exponent $-0.55$ (solid line) lies between $-0.5$ (pure mean-field dilution, dashed) and $-1.0$ (fully correlated systems, dotted), reflecting moderate topological correlation in generator coupling. Although per-generator impact decreases with $N$, the aggregate erosion $n_g \times (N^{-0.55}$ term$) = N^{0.45}$ grows superlinearly, explaining why large grids show nearly double the margin gap of small grids despite smaller per-generator contributions.

consistent model ranking. All comparisons remain significant at $p < 0.001$ after Holm-Sidak correction for six simultaneous tests.

### *F. Stressed False-Safety Analysis*

To assess whether the nominal margin gap extends to realistic adverse conditions, we evaluate all models under a stress grid spanning packet-loss rates $p \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ and jitter standard deviations $\sigma_j \in \{20, 40, 60, 80\}$ ms (26 stress levels $\times$ 5 seeds per case). Under maximum stress ($p = 0.30$, $\sigma_j = 80$ ms), B10 reports margins of 0.026, 0.038, and 0.058 for Case-39, Case-118, and PEGASE-1354, respectively, identical to its nominal values since it lacks $\Delta_R$ and $\Delta_J$ terms. In contrast, the proposed method reports reduced margins of 0.007, 0.010, and 0.016 for the same systems, yielding gap ratios of 3.6×, substantially higher than the nominal 11–25% (1.11–1.25×) reported in Table V. Although no false-safe events (B10 stable, Ours unstable) occur at these stress levels, indicating that genuine instability requires impairments beyond $p = 0.30$ or $\sigma_j = 80$ ms, the absence of false-safe events does not imply the gap is benign: the 3.6× gap means that operators relying on delay-only margins would allocate safety reserves that are in reality already partially consumed by unmodeled erosion. In operational terms, this gap demonstrates that the delay-only bound would substantially under-provision safety margins in cellular backhaul links where $p = 0.10$–$0.15$ and $\sigma_j = 40$–$60$ ms are common [29], even when outright instability is not triggered.

### G. Hyperparameter Sensitivity and Erosion Budget

TABLE VII
HYPERPARAMETER SENSITIVITY: $\rho$ SENSITIVITY (RELATIVE RANGE) TO EACH HYPERPARAMETER ACROSS THREE REPRESENTATIVE CASES. THE LOSS WEIGHT $\alpha$ AND NORMALIZATION COEFFICIENT $\beta$ ARE ROBUST ($\leq 3.4\%$); THE MARGIN SCALE $s$ AND BUDGET SPLIT SHOW MODERATE SENSITIVITY (8–13%).

| **Parameter** | **C-39** $\downarrow$ | **C-118** $\downarrow$ | **P-1354** $\downarrow$ |
|---|---|---|---|
| $\alpha \in [0.1, 5.0]$ | 3.4% | 1.2% | 0.6% |
| $\beta \in [0.1, 2.0]$ | 3.4% | 1.2% | 0.6% |
| $s \in [0.7, 0.95]$ | 12.9% | 11.3% | 10.6% |
| Budget split | 10.9% | 9.2% | 8.4% |
| *Empirical erosion fractions (mean across all sweeps):* | | | |
| $\Delta_K$ (delay) | 41.9% | 35.7% | 35.2% |
| $\Delta_R$ (packet loss) | 50.7% | 57.9% | 58.2% |
| $\Delta_J$ (jitter) | 7.4% | 6.4% | 6.6% |

Table VII quantifies the sensitivity of the stability margin to each hyperparameter and reports the empirical erosion decomposition. The regularization weights $\alpha$ and $\beta$ produce less than 3.4% variation in $\rho$ across a 50$\times$ range; this robustness is expected because $\alpha$ and $\beta$ scale auxiliary loss terms ($\mathcal{L}_{\text{couple}}$, $\mathcal{L}_{\text{channel}}$) whose gradients are small relative to the dominant $\mathcal{L}_{\text{OPF}}$, so moderate changes in their magnitude do not substantially alter the learned coupling constants. The margin scale $s$ and budget allocation show moderate sensitivity of 8–13%, consistent with their role in directly setting the initial coupling constant magnitudes via (18); these results support using the recommended defaults ($\alpha = 1.0$, $\beta = 0.5$, $s = 0.9$, 50/30/20 budget split) without per-grid tuning. Furthermore, the empirical erosion fractions reveal that packet-loss erosion $\Delta_R$ accounts for 50.7–58.2% of the total erosion, consistently exceeding the delay contribution $\Delta_K$ (35–42%), while jitter $\Delta_J$ contributes only 6–7%. The packet-loss dominance over jitter ($> 88\%$ of the combined $\Delta_R + \Delta_J$ budget) confirms that the budget allocation of 30% to packet loss and 20% to jitter is well-calibrated. The learnable $R_i$ constants absorb the excess packet-loss erosion beyond the initial budget allocation, allowing the bound to remain tight without manual retuning.

### H. Burst-Loss Model Robustness

Table VIII evaluates whether the i.i.d. Bernoulli loss model in Theorem 2 remains a valid bound when the true channel exhibits bursty Gilbert-Elliott (GE) behavior. For each target mean loss rate $p \in \{0.05, 0.10, 0.20\}$, GE parameters are fitted to match the target $p$ via the steady-state burst probability $\pi_B = q_{gb}/(q_{bg}+q_{gb})$, where $q_{bg}$ and $q_{gb}$ are the BAD$\rightarrow$GOOD and GOOD$\rightarrow$BAD transition rates with $p_{\text{bad}} = 0.40$ and $p_{\text{good}} = 0.01$. The key finding is that the GE channel erodes the stability margin by only 1.7–3.2% more than the i.i.d. channel at matched mean loss rates. This small discrepancy arises because the Bauer-Fike perturbation bound depends on the expected norm of the holding-time perturbation, which is determined primarily by the mean loss rate $p$ rather than the temporal correlation structure; bursty channels produce longer individual holding episodes but fewer of them per unit time, yielding a similar expected perturbation. The i.i.d. bound

TABLE VIII
GILBERT-ELLIOTT VS. I.I.D. BERNOULLI BURST-LOSS COMPARISON: STABILITY MARGIN $\rho$ AND EROSION TERM $\Delta_R$ RATIOS AT MATCHED MEAN LOSS RATES ACROSS THREE REPRESENTATIVE CASES AND 5 SEEDS. GE ERODES THE MARGIN BY ONLY 1.7–3.2% MORE THAN I.I.D. AT MATCHED RATES, CONFIRMING THAT THE I.I.D. BERNOULLI BOUND IS MILDLY CONSERVATIVE.

| **Case** | $p$ | **GE** $\rho \uparrow$ | **i.i.d.** $\rho \uparrow$ | $\Delta_R$ **ratio** $\approx$**1** |
|---|---|---|---|---|
| Case-39 | 0.05 | 0.0273 | 0.0274 | 1.017 |
| | 0.10 | 0.0247 | 0.0248 | 1.032 |
| | 0.20 | 0.0186 | 0.0188 | 1.024 |
| Case-118 | 0.05 | 0.0340 | 0.0341 | 1.025 |
| | 0.10 | 0.0309 | 0.0310 | 1.024 |
| | 0.20 | 0.0240 | 0.0242 | 1.018 |
| PEGASE-1354 | 0.05 | 0.0517 | 0.0519 | 1.023 |
| | 0.10 | 0.0467 | 0.0470 | 1.025 |
| | 0.20 | 0.0364 | 0.0368 | 1.017 |

$\Delta_R = \sum_i R_i \cdot p/(1-p)$ therefore acts as a conservative but tight approximation under burst loss. This mildly conservative property is desirable for safety-critical applications: the i.i.d. bound will never *under*-estimate the erosion induced by bursty real-world channels, while the excess conservatism of 1.7–3.2% is negligible compared to the 11–25% margin gap of delay-only models. Moreover, the GE mean run length of 0.05–0.25 packets confirms that burst durations in the tested operating range are short relative to the control period, justifying per-packet mean loss as the primary metric.

Taken together, the eight experiments establish three key conclusions: (1) the margin gap between delay-only and multi-impairment bounds is 11–25%, widening to 3.6$\times$ under stressed impairments; (2) the proposed bound is robust to hyperparameter choices and bursty channel conditions; and (3) the framework scales gracefully from 39-bus regional grids to 2,869-bus pan-European systems with negligible parameter and latency overhead.

## VII. LIMITATIONS AND PRACTICAL APPLICATIONS

While the results of Section VI demonstrate consistent improvements across all eight benchmarks, several assumptions constrain the generality of the current framework. This section identifies these limitations transparently and then outlines the practical deployment scenarios in which the proposed approach is most immediately applicable.

### A. Limitations

Four limitations bound the scope of the present results. First, Theorem 2 relies on linearization around a stable equilibrium (Assumption A1), which is valid in the small-signal regime but does not capture large-deviation dynamics during fault-induced transients or post-contingency recovery. Extensions to nonlinear Lyapunov-Krasovskii functionals that preserve the packet-loss and jitter erosion structure remain an open problem.

Second, the additive decomposition via the triangle inequality is inherently conservative when the three perturbation

mechanisms are correlated or partially cancel through favorable grid topology. The learnability of $K_i$, $R_i$, $J_i$ mitigates this conservatism from data but cannot eliminate it analytically; the bound remains a valid lower bound on the true margin regardless.

Third, cross-impairment interaction terms ($K_iR_j$, $K_iJ_j$, $R_iJ_j$) are absent from the current formulation, which assumes independent perturbation mechanisms at each generator's communication channel. Correlated impairments arising from shared network congestion sources or common propagation paths may require an extended formulation with multiplicative coupling terms.

Fourth, the largest evaluated system contains 2,869 buses, while real-world continental transmission grids may exceed 10,000 buses. Extending the architecture to that scale will require sparse attention mechanisms and distributed GNN aggregation to maintain computational tractability. Future work should address time-varying impairment profiles (non-stationary loss bursts, weather-driven jitter spikes) and include hardware-in-the-loop validation on OPAL-RT platforms with live PMU data streams to establish field-condition deployment readiness.

### B. Practical Applications

The proposed framework is readily applicable to three classes of grid operational infrastructure. **Energy management system (EMS) security assessment**: the ResilientJointOptimizer can replace or augment the delay-only stability check currently embedded in EMS contingency screening, providing operators with a corrected margin $\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j)$ that accounts for the actual channel state. The 11–25% overestimation quantified in Section VI-C corresponds to the safety buffer that EMS operators may unknowingly consume when running delay-only assessments; correcting it prevents the system from being dispatched into regions that appear safe but are not.

**Communication infrastructure planning**: the 88/12 decomposition of non-delay erosion (packet loss dominant) provides a quantitative basis for prioritizing communication investment. Forward error correction, retransmission protocol tuning, and loss-aware routing address $\Delta_R$, the larger term, while playout buffering and traffic shaping address $\Delta_J$. The $N^{-0.55}$ scaling law further informs planning for grid expansion: each tenfold increase in system size increases aggregate non-delay erosion by approximately $10^{0.45} \approx 2.8\times$, requiring proportionally more aggressive loss-mitigation measures in the communication overlay.

**Real-time SCADA and PMU integration**: the sub-60 ms inference latency across all eight evaluated cases satisfies IEEE C37.118.1 PMU reporting requirements, and the 2.9–3.8% parameter overhead relative to the delay-only model means the ResilientJointOptimizer can be deployed on existing substation automation hardware without additional compute infrastructure. Per-generator coupling constants $K_i$, $R_i$, $J_i$ can be periodically retrained on rolling windows of operational PMU data, enabling adaptive stability bounds that track seasonal load patterns, renewable intermittency cycles, and long-term communication infrastructure degradation.

Despite the limitations identified above, the combination of formal multi-impairment guarantees, negligible computational overhead, and compatibility with existing SCADA infrastructure positions the proposed framework for near-term deployment. The next section summarizes the principal findings and outlines directions for future work.

## VIII. Conclusion

Addressing the gap between delay-only stability theory and the multi-impairment reality of deployed communication networks, this paper presented a unified framework centered on Theorem 2, a formal lower bound on the stability margin under simultaneous delay, packet loss, and jitter with learnable per-generator coupling constants $K_i$, $R_i$, $J_i$. The theorem decomposes total margin erosion into three additive terms derived from independent physical mechanisms: Padé-based delay perturbation, Bernoulli geometric-distribution packet-loss holding, and second-order zero-mean jitter variance, combined via Bauer-Fike eigenvalue perturbation and the triangle inequality. The ResilientJointOptimizer learns all three coupling constant families end-to-end via gradient descent with budget-aware initialization, providing the most accurate multi-impairment margin estimates across eight real power systems spanning 39 to 2,869 buses. Four principal findings emerge. First, the margin gap between delay-only and multi-impairment bounds is 11–12% on small/medium grids and 24–25% on bulk grids; under stressed impairments ($p = 0.30$, $\sigma_j = 80\,\mathrm{ms}$), the gap ratio grows to $3.6\times$. Second, learnable coupling constants reduce conservatism relative to fixed worst-case constants by 20–30%, confirming that data-driven calibration is essential. Third, packet loss accounts for $\sim$88% of non-delay erosion and jitter for $\sim$12%, providing a quantitative priority ordering for communication infrastructure investment. Fourth, per-generator erosion follows a $N^{-0.55}$ power law, explaining the near-doubling of the margin gap between small-grid and large-grid regimes.

## Appendix A
## Proof of Theorem 2 (Multi-Impairment Stability Bound)

This appendix provides the complete proof of Theorem 2. The proof extends the delay-only Theorem 1 of [24] by adding two new perturbation bounding steps—packet loss via Bernoulli sample-and-hold analysis and jitter via second-order stochastic Taylor expansion—and assembles all three via the triangle inequality and the Bauer-Fike eigenvalue perturbation theorem.

### A.1 Problem Setup and Total Perturbation Decomposition

Consider the linearized impaired system (2). The three channel impairments affect the system Jacobian through independent perturbation mechanisms that can be treated separately before combination. The total perturbation to the system matrix is

$$\Delta\mathbf{J}_{\text{total}} = \Delta\mathbf{J}_{\tau} + \Delta\mathbf{J}_{\text{loss}} + \Delta\mathbf{J}_{\text{jitter}}, \tag{22}$$

where $\Delta\mathbf{J}_\tau$ is the delay-induced perturbation, $\Delta\mathbf{J}_{\text{loss}}$ is the packet-loss-induced perturbation, and $\Delta\mathbf{J}_{\text{jitter}}$ is the jitter-induced perturbation. The additive structure in (22) is justified by the independence of the three mechanisms: delay acts through a time shift of the control input, packet loss acts through a sample-and-hold substitution when packets are dropped, and jitter acts through a stochastic variance perturbation of the effective delay. No cross-mechanism interactions (e.g., simultaneous loss and jitter at the same time step) produce terms beyond the individual contributions at the level of the first-order Padé and second-order Taylor approximations used in the derivation.

### *A.2 Step 1: Delay Perturbation via Padé Approximation*

Under Assumption (A4), the delay operator $e^{-\tau_i s}$ is replaced by the first-order $[1/1]$ Padé approximant from (5). This converts the delay-differential equation in (2) into a perturbed ordinary differential equation with an additive perturbation to the system matrix. A first-order Taylor expansion in the delay variables $\tau_i$ and the triangle inequality give

$$\|\Delta\mathbf{J}_\tau\|_2 \;\le\; \sum_{i=1}^{n_g} \left\| \frac{\partial \mathbf{J}}{\partial \tau_i} \right\|_2 \cdot \tau_i + O\big(\|\boldsymbol{\tau}\|^2\big)\,. \tag{23}$$

Normalizing each delay by its maximum and defining the delay coupling constant as $K_i := \mathrm{cond}(\mathbf{V}) \cdot \|\partial\mathbf{J}/\partial\tau_i\|_2 \cdot \tau_{\max,i}$ (where the condition number $\mathrm{cond}(\mathbf{V})$ of the eigenvector matrix is absorbed into the learnable constant, as detailed in Section A.3), the first-order term yields the delay erosion $\|\Delta\mathbf{J}_\tau\|_2 \le \Delta_K = \sum_i K_i \cdot \tau_i/\tau_{\max,i}$, reproducing the delay-only bound of Theorem 1 [24].

**Padé approximation error.** The Taylor series of $e^{-\tau s}$ and of the $[1/1]$ Padé approximant $(1-\tau s/2)/(1+\tau s/2)$ agree through the quadratic term; their difference is

$$e^{-\tau s} - \frac{1-\tau s/2}{1+\tau s/2} \;=\; \frac{(\tau s)^3}{12} + O\big((\tau s)^4\big)\,. \tag{24}$$

The Padé residual in (24) therefore introduces an additional perturbation of norm at most $|\tau_i|^3|s|^3/12$ per generator. In the typical operating regime—delay mean 50 ms, electromechanical mode frequencies $|s| \lesssim 5\,\mathrm{rad/s}$—the product $|\tau_i s| \approx 0.25$, yielding a residual of at most $(0.25)^3/12 \approx 1.3\times10^{-3}$, negligible relative to the stability margins of 0.03–0.08 reported in Table IX. Near the upper bound $\tau_{\max} = 500\,\mathrm{ms}$, the product $|\tau_i s|$ can reach 2.5 and the Padé residual grows to $\approx 1.3$; in this regime the $[1/1]$ Padé overestimates the true delay perturbation, making the bound conservative. The learnable coupling constants $K_i$ correct for this conservatism by calibrating against the actual observed stability degradation via $\mathcal{L}_{\text{couple}}$, effectively incorporating a data-driven Padé correction without requiring an explicit higher-order expansion.

### *A.3 Step 2: Eigenvalue Perturbation via Bauer-Fike Theorem*

Under Assumption (A3), the zero-delay Jacobian is diagonalizable as $\mathbf{J}(0) = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}$. The Bauer-Fike theorem [34] states that for any perturbation matrix $\mathbf{E}$ and any eigenvalue $\tilde{\lambda}$ of $\mathbf{J}(0)+\mathbf{E}$, there exists an eigenvalue $\lambda_k(0)$ of $\mathbf{J}(0)$ such that

$$|\tilde{\lambda} - \lambda_k(0)| \;\le\; \mathrm{cond}(\mathbf{V}) \cdot \|\mathbf{E}\|_2. \tag{25}$$

This bound links the norm of the total perturbation matrix to the maximum shift in any eigenvalue, including the least stable eigenvalue $\lambda_{\min}(0)$. The condition number $\mathrm{cond}(\mathbf{V}) = \|\mathbf{V}\|_2\|\mathbf{V}^{-1}\|_2 \ge 1$, with equality when $\mathbf{J}(0)$ is normal.

**Role of $\mathrm{cond}(\mathbf{V})$ and its absorption into coupling constants.** The condition number $\mathrm{cond}(\mathbf{V})$ depends on the eigenvector structure of $\mathbf{J}(0)$, which changes with the operating point through the synchronizing torque matrix $\mathbf{S} = \partial\mathbf{P}_e/\partial\boldsymbol{\theta}|_{\boldsymbol{\theta}_0}$. Computing $\mathrm{cond}(\mathbf{V})$ analytically requires accurate knowledge of the full generator parameter set and network admittance matrix at each operating point—information that is rarely available with the precision needed for a tight bound. Rather than propagating a single worst-case $\mathrm{cond}(\mathbf{V})$ through the bound (which would introduce a potentially large and conservative constant), the proposed framework defines $K_i$, $R_i$, $J_i$ to *include* $\mathrm{cond}(\mathbf{V})$ as a multiplicative factor. The learnable constants therefore implicitly track the operating-point-dependent eigenvector sensitivity without requiring its explicit computation. This is not an approximation: for any fixed operating point, an exact $\mathrm{cond}(\mathbf{V})$ exists and is finite under Assumption (A3), so the coupling constants are well-defined. Across operating points sampled during training, gradient descent on $\mathcal{L}_{\text{couple}}$ drives $K_i$, $R_i$, $J_i$ toward values that are consistent with the *distribution* of $\mathrm{cond}(\mathbf{V})$ encountered in practice, producing a tighter and more realistic bound than any single worst-case value.

### *A.4 Step 3: Packet-Loss Perturbation*

Under Assumption (A5), packet arrivals at generator $i$ follow an independent Bernoulli process with success probability $1-p_i$. When a packet is lost, the controller applies a sample-and-hold strategy: the last successfully received setpoint is maintained until the next successful reception. The derivation of $\Delta_R$ proceeds in three sub-steps.

**Sub-step 3a: Geometric mean holding time.** The number of consecutive losses before a successful reception follows a geometric distribution $N_{\text{hold}} \sim \mathrm{Geom}(1-p_i)$, with probability mass function $\Pr[N_{\text{hold}} = k] = p_i^k(1-p_i)$ for $k = 0, 1, 2, \ldots$ The expected additional holding time per loss event is therefore

$$\mathbb{E}\big[T_{\text{hold},i}\big] \;=\; T_s \cdot \mathbb{E}[N_{\text{hold}}] \;=\; T_s \cdot \frac{p_i}{1-p_i}, \tag{26}$$

where $T_s$ is the sampling period. The expected holding time in (26) acts as an effective increment to the communication delay experienced by generator $i$'s control input.

**Sub-step 3b: Perturbation norm bound.** Treating the expected holding time from (26) as an additional delay increment $\delta\tau_i = T_s \cdot p_i/(1-p_i)$ and applying the same first-order Padé linearization from Step 1 (equation (23)) to this increment, the induced perturbation norm for generator $i$ satisfies

$$\|\Delta\mathbf{J}_{\text{loss},i}\|_2 \;\le\; \left\| \frac{\partial \mathbf{J}}{\partial \tau_i} \right\|_2 \cdot \delta\tau_i \;=\; \left\| \frac{\partial \mathbf{J}}{\partial \tau_i} \right\|_2 \cdot T_s \cdot \frac{p_i}{1-p_i}. \tag{27}$$

**Sub-step 3c: Assembly with Bauer-Fike.** Defining the packet-loss coupling constant as $R_i := \mathrm{cond}(\mathbf{V}) \cdot \|\partial \mathbf{J}/\partial \tau_i\|_2 \cdot T_s$ (which absorbs both the eigenvector condition number and the sampling period into the learnable scalar) and applying the triangle inequality over all generators yields the per-generator bound from (27) summed as

$$\|\Delta \mathbf{J}_{\mathrm{loss}}\|_2 \;\le\; \sum_{i=1}^{n_g} R_i \cdot \frac{p_i}{1-p_i} \;=\; \Delta_R, \tag{28}$$

where $R_i$ in (28) captures the per-generator sensitivity of the system matrix to the geometric-mean holding delay. The functional form $p/(1-p)$ is exact under the Bernoulli loss model: it grows sublinearly for small $p$, equals 1 at $p = 0.5$, and diverges as $p \to 1$, correctly reflecting the physical impossibility of maintaining stable control when the channel is almost always silent. Assumption (A5) requires $p_i < p_{\mathrm{crit}} = 0.6$ to ensure the bound remains finite; empirically, packet loss rates above 30% are rare in operational grid communication links [29].

### *A.5 Step 4: Jitter Perturbation via Second-Order Stochastic Analysis*

Jitter introduces a zero-mean stochastic perturbation $\xi_i(t)$ to the nominal delay $\tau_i$, with $\mathbb{E}[\xi_i] = 0$ and $\mathrm{Var}[\xi_i] = \sigma_{j,i}^2$ per Assumption (A5). The effective delay at time $t$ is $\tau_i + \xi_i(t)$. Expanding the Jacobian as a function of the effective delay to second order in $\xi_i$ gives the Taylor expansion

$$\mathbf{J}(\tau_i + \xi_i) = \mathbf{J}(\tau_i) + \frac{\partial \mathbf{J}}{\partial \tau_i}\xi_i + \frac{1}{2}\frac{\partial^2 \mathbf{J}}{\partial \tau_i^2}\xi_i^2 + O(\xi_i^3). \tag{29}$$

Taking expectations over the jitter distribution in (29) and using $\mathbb{E}[\xi_i] = 0$, $\mathbb{E}[\xi_i^2] = \sigma_{j,i}^2$ yields

$$\mathbb{E}[\mathbf{J}(\tau_i + \xi_i)] = \mathbf{J}(\tau_i) + \frac{1}{2}\frac{\partial^2 \mathbf{J}}{\partial \tau_i^2}\sigma_{j,i}^2 + O(\sigma_{j,i}^3). \tag{30}$$

The first-order term vanishes *identically* because $\mathbb{E}[\xi_i] = 0$; this is the zero-mean assumption of Assumption (A5) invoked explicitly here. The dominant contribution to the expected perturbation in (30) is therefore the second-order term, proportional to the jitter variance $\sigma_{j,i}^2$. Bounding the operator norm and normalizing by the maximum jitter variance $\sigma_{\max,i}^2$ gives the jitter erosion

$$\|\Delta \mathbf{J}_{\mathrm{jitter}}\|_2 \;\le\; \sum_{i=1}^{n_g} J_i \cdot \frac{\sigma_{j,i}^2}{\sigma_{\max,i}^2} \;=\; \Delta_J, \tag{31}$$

where $J_i := \frac{1}{2}\,\mathrm{cond}(\mathbf{V}) \cdot \|\partial^2 \mathbf{J}/\partial \tau_i^2\|_2 \cdot \sigma_{\max,i}^2$. The quadratic dependence of $\Delta_J$ in (31) on the jitter standard deviation $\sigma_{j,i}$ (rather than linear) is a fundamental consequence of zero-mean jitter: the first-order stability effect averages out exactly, and the dominant degradation comes entirely from the variance of the delay fluctuation. This explains why reducing jitter standard deviation by a factor of 2 reduces the jitter erosion by a factor of 4, making jitter suppression efficient once the variance is already small.

### *A.6 Step 5: Triangle Inequality Assembly and Final Bound*

By (22) and the triangle inequality applied to the three independent perturbation norms:

$$\begin{aligned}\|\Delta \mathbf{J}_{\mathrm{total}}\|_2 \;&\le\; \|\Delta \mathbf{J}_{\tau}\|_2 + \|\Delta \mathbf{J}_{\mathrm{loss}}\|_2 + \|\Delta \mathbf{J}_{\mathrm{jitter}}\|_2 \\ &\le\; \Delta_K + \Delta_R + \Delta_J.\end{aligned} \tag{32}$$

The absence of cross-terms ($K_iR_j$, $K_iJ_j$, $R_iJ_j$) in (32) is justified by the independence of the three perturbation mechanisms: delay, sample-and-hold loss, and jitter variance act through separate channels and independent random processes at each generator's communication link, so their combined Jacobian perturbation is the sum of the individual contributions without multiplicative interaction terms at the level of the approximations used.

Combining (32) with the Bauer-Fike bound (25) for the total perturbation $\mathbf{E} = \Delta \mathbf{J}_{\mathrm{total}}$, every eigenvalue $\lambda_k(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j)$ of the impaired system satisfies

$$|\lambda_k(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j) - \lambda_k(0)| \;\le\; \mathrm{cond}(\mathbf{V}) \cdot (\Delta_K + \Delta_R + \Delta_J). \tag{33}$$

The stability margin $\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j) = -\max_k \mathrm{Re}(\lambda_k(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j))$ is bounded below by applying (33) to the least stable mode $\lambda_{\min}(0)$:

$$\rho(\boldsymbol{\tau}, \mathbf{p}, \boldsymbol{\sigma}_j) \;\ge\; |\lambda_{\min}(0)| - (\Delta_K + \Delta_R + \Delta_J), \tag{34}$$

which is precisely the bound stated in Theorem 2, equations (7)–(6). This completes the proof. ■

### *A.7 Remark: Tightness, Learnability, and Validity Domain*

The bound (34) becomes tight under three conditions: (i) the three perturbations are rank-1 and aligned with the dominant eigenvector of $\lambda_{\min}(0)$; (ii) the system Jacobian $\mathbf{J}(0)$ is normal, so $\mathrm{cond}(\mathbf{V}) = 1$; and (iii) the Padé approximation error and the higher-order Taylor terms are negligible. In practice, at least one of these conditions is violated, making the bound conservative. The learnability of $K_i$, $R_i$, $J_i$ addresses this conservatism: by minimizing the coupling loss $\mathcal{L}_{\mathrm{couple}} = \|\rho_{\mathrm{emp}} - \rho_{\mathrm{theo}}\|_2^2$ during training, the constants adapt to the actual observed stability degradation rather than worst-case analytical values, effectively incorporating partial cross-term cancellation and Padé correction through data without requiring an explicit second-order extension.

**Quantitative bound on triangle inequality conservatism.** The triangle inequality in (32) is tight only when all three perturbation norms are aligned. When the three terms $\Delta_K$, $\Delta_R$, $\Delta_J$ are mutually orthogonal in the matrix-norm sense—which holds when delay, packet-loss, and jitter act on different subspaces of the generator state space—the true combined norm satisfies

$$\begin{aligned}\|\Delta \mathbf{J}_{\mathrm{total}}\|_2 \;&=\; \sqrt{\Delta_K^2 + \Delta_R^2 + \Delta_J^2} \\ &\le\; \Delta_K + \Delta_R + \Delta_J \\ &\le\; \sqrt{3}\,\|\Delta \mathbf{J}_{\mathrm{total}}\|_2.\end{aligned} \tag{35}$$

The right-hand inequality in (35) shows that the triangle inequality overestimates the true perturbation by at most a factor of $\sqrt{3} \approx 1.73$ when all three terms are equal. In

TABLE IX
POWER SYSTEM BENCHMARKS

| System | $N$ | $n_g$ | $\lvert\lambda_{\min}(0)\rvert$ | Real-World Analog and Source |
|---|---|---|---|---|
| Case-39 [39] | 39 | 10 | 0.035 | New England; 10-machine transient stability benchmark |
| Case-57 [39] | 57 | 7 | 0.042 | US Midwest grid (partial); medium-scale standard case |
| Case-118 [39] | 118 | 54 | 0.051 | Interconnected regional grid; 54-unit algorithmic benchmark |
| Case-145 [39] | 145 | 50 | 0.077 | Upper Midwest grid; 50-unit system with dense tie-line structure |
| Case-300 [39] | 300 | 69 | 0.061 | Eastern US interconnection; largest standard test case |
| PEGASE-1354 [42] | 1,354 | 260 | 0.077 | European meshed grid; PEGASE continental planning study |
| RTE-1888 [43] | 1,888 | 296 | 0.072 | French national transmission; real RTE operational snapshot |
| PEGASE-2869 [42] | 2,869 | 510 | 0.079 | Pan-European grid; largest benchmark, 510 generators |

practice, the budget allocation $\Delta_K : \Delta_R : \Delta_J \approx 5 : 3 : 2$ (50/30/20%) gives an overestimation factor of at most $(\Delta_K + \Delta_R + \Delta_J)/\sqrt{\Delta_K^2 + \Delta_R^2 + \Delta_J^2} \leq (5+3+2)/\sqrt{25+9+4} = 10/\sqrt{38} \approx 1.62$. The learnable coupling constants absorb this systematic conservatism during training: since $\mathcal{L}_{\text{couple}}$ penalizes overestimation of $\rho_{\text{emp}}$ by $\rho_{\text{theo}}$, gradient descent implicitly drives the constants toward values that compensate for the triangle inequality gap.

**Empirical tightness on small cases.** For the three smallest cases (Case-39, Case-57, Case-118), we verify tightness by comparing the theoretical bound $\rho_{\text{theo}} = |\lambda_{\min}(0)| - \Delta_K - \Delta_R - \Delta_J$ to the empirical stability margin $\rho_{\text{emp}}$ computed by direct numerical eigenvalue analysis of the perturbed linearized system at 100 operating points per case. The mean ratio $\rho_{\text{theo}}/\rho_{\text{emp}}$ is $0.91 \pm 0.03$ for Case-39, $0.89 \pm 0.04$ for Case-57, and $0.87 \pm 0.05$ for Case-118, confirming that the bound is within 9–13% of the true margin and therefore sufficiently tight for operational safety assessments.

The validity domain of the bound is $0 \leq p_i < 1$ and $\sigma_{j,i} \geq 0$ for all $i$, with the tightening conditions ensuring $\rho > 0$ (stability) throughout the operating regime. The theoretical bound is confirmed empirically across all eight test systems in Section VI.

## APPENDIX B
## EXPERIMENTAL CONFIGURATION DETAILS

This appendix provides the power system benchmarks and the complete hyperparameter configuration used in all experiments.

### A. Hyperparameter Configuration

Table X lists all architecture, training, and loss function hyperparameters. The embedding dimension $d = 128$ and the coupling weight $\alpha = 1.0$ are validated by the sensitivity analysis in Section VI-G. The budget-aware initialization computes $K_{\text{init}}$ from the system eigenvalue structure according to (18).

### B. Impairment Distributions

For each of the eight benchmark systems, 1,000 operating scenarios are generated. Communication impairments are sampled independently per generator: delays from a truncated lognormal ($\mu = 50\,\text{ms}$, $\sigma = 20\,\text{ms}$, $\tau \in [5, 500]\,\text{ms}$), packet loss rates from a $\text{Beta}(\alpha{=}1, \beta{=}19)$ distribution (mean 5%, range $[0, 30\%]$), and jitter standard deviations from a $\text{Gamma}(k{=}4, \theta{=}2.5)$ distribution (mean 10 ms, range $[0, 50]\,\text{ms}$). SNR and buffer size are drawn uniformly from $[5, 30]\,\text{dB}$ and $[10, 100]$ packets, respectively. The data split is 70/15/15% (train/val/test) with fixed random seeds for reproducibility.

TABLE X
COMPLETE HYPERPARAMETER CONFIGURATION.

| Category | Parameter | Value |
|---|---|---|
| **Architecture** | Embedding dim $d$ | 128 |
| | Hidden dim $d_{\text{hidden}}$ | 256 |
| | GNN layers $L$ | 3 |
| | GNN type | GAT |
| | Attention heads $H$ | 8 |
| | Attention dropout | 0.1 |
| | Energy input dim | 5 $(P, Q, V, \theta, \omega)$ |
| | Comm. input dim | 6 $(\tau, p, \sigma_j, R_{\text{retx}}, B, \text{SNR})$ |
| **Training** | Optimizer | AdamW |
| | Learning rate | $1 \times 10^{-4}$ |
| | Weight decay | $1 \times 10^{-5}$ |
| | Scheduler | Cosine w/ 10-epoch warmup |
| | Min learning rate | $1 \times 10^{-6}$ |
| | Gradient clipping | Norm 1.0 |
| | Batch size | 32 |
| | Early stopping patience | 50 epochs |
| **Loss** | Cost weight | 1.0 |
| | Voltage weight | 10.0 |
| | Frequency weight | 100.0 |
| | Coupling weight $\alpha$ | 1.0 |
| | Channel weight $\beta$ | 0.5 |
| | Stability hinge $\rho_{\min}$ | 0.01 |
| **Stability** | $K/R/J$ init | Budget-aware: $s \cdot \lvert\lambda_{\min}\rvert/n_g$ |
| | Safety factor $s$ | 0.9 |
| | Budget split $K : R : J$ | 50 : 30 : 20 |
| | $\tau_{\max}$ | 500 ms |
| | Channel states | 3 (GOOD / DEGRADED / FAILED) |
| **Stats.** | Seeds | 20 $(\{0, 1, \ldots, 19\})$ |
| | Primary test | Wilcoxon signed-rank |
| | Correction | Holm-Sidak |
| | Significance level | $p < 0.001$ |

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